Supplementary Information: Reconstructing founder haplotypes, pedigrees and recombination from low-coverage sequencing of experimental crosses

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This supplement gives the finer rules of every stage, with the default value of each named parameter. Codes in parentheses (for example EC05 or AJ17) identify the method unit in the software's methods documentation (`docs/methods/` in the repository), where each rule is stated in full together with the code that implements it. Default values are those of the generated parameter reference (`docs/reference/parameters.md`) for release 4.4.0.

S1. Read evidence, quality control and relatedness

Input admission and the sample axis

A record is admitted when it has one ALT allele and both alleles are single nucleotides (A, C, G, T). All admitted records sharing a position are removed before AD is decoded. Missing AD gives zero reads. Input FILTER is not a quality decision (EC01). The sample axis is sorted by name, with the same permutation applied to every per-sample array, and outputs are restored to input order (EC03).

Read-model fitting

Folds (EC17). Per-sample (depth, ALT) histograms of cells with $1 \le d \le$ `maximum_fit_depth` = 48 are kept separately for training cells (fold $\lfloor x_j / w_\mathrm{f} \rfloor \bmod 4 \in \{0, 1\}$, w_f = `fold_window_bp` = 1 Mb) and held-out cells.

Shared binomial (EC18). EM starts from $(e, q) \in$ `em_starts` = {(0.02, 0.5), (0.01, 0.35), (0.08, 0.65)} with genotype weights 1/3; pooled M-steps set e and q to expected wrong-allele (heterozygote ALT) reads over expected homozygote (heterozygote) depth, requiring $0 < e < q < 1 - e$.

EM stops when the gain is at most `em_tolerance_per_observation` = 10^{-10} per observation. A start fails at `em_maximum_iterations` = 2,000, or on a likelihood decrease of more than 10^{-7} per observation. Each endpoint is then Newton-polished to stationarity in at most 40 box-constrained steps.

Endpoints are equivalent within relative distances of 10^{-13} (shared logits) and 10^{-8} (nuisance logits). The representative endpoint has the largest training log-likelihood; ties go to the lexicographically smallest parameters. A distinct endpoint within 1 nat of the best makes the fit unresolved. A failed polish fails its start; an unresolved fit, or failure of every start, gives the fixed model.

Pooled sample binomial (EC19). Each sample is first fitted without a prior, by L-BFGS-B (Byrd et al. 1995) plus a Newton polish. The variances v_i of its two logits are diagonal entries of the inverse observed Hessian, so its genotype mixture is profiled out. They are used only if the smallest Hessian eigenvalue exceeds 10^{-8}. For each logit y, (μ, τ) minimise

$$\tfrac{1}{2}\sum_i [\log(v_i + \tau) + \frac{(y_i - \hat{\mu}(\tau))^2}{v_i + \tau}], \qquad 0 \le \tau \le \max\{1, 4\,\mathrm{var}(y)\},$$

where $\hat{\mu}(\tau)$ is the precision-weighted mean. The minimiser is refined to a root of the derivative. $\tau = 0$ is kept when its profile value lies within a relative 10^{-10} of the optimum.

With fewer than three finite variances, μ takes the shared value and $\tau = 0$. Samples are refitted under $\mathcal{N}(\mu, \tau)$; a $\tau = 0$ effect is fixed at μ, and a sample without training reads takes the prior mode.

Joint beta-binomial (EC20). The fit starts from the pooled effects with κ = `initial_heterozygote_concentration` = 40, under the pooled prior. L-BFGS-B is followed by a Newton polish on the arrowhead Hessian: one 4×4 block per sample, coupled to $\log\kappa$. The bounds are $\mathrm{logit}\, e \in [-16, -10^{-8}]$, $\mathrm{logit}\, b \in [-12, 12]$, $\log\kappa \in [-8, 16]$ and mixing logits in $[-25, 25]$.

Selection (EC21–EC22). Stretches without admitted cells are dropped. Complex candidates with non-finite held-out totals are skipped, and a failed complex fit never discards a valid simpler one. Acceptance needs $n_h \ge$ `selection_minimum_blocks` = 4, $\Delta > 0$ and $z \ge$ `selection_z` = 2 (inclusive), and a rejection does not stop later comparisons. The fallback model is binomial, with `fixed_error` = 0.02 and balance 1/2.

Application (EC16, EC23). Emissions are exact integer-depth rising products, exact in the binomial limit $1/\kappa = 0$. The `lookup_depth` = 64 table changes cost only. Validation requires exactly 1/3 at zero depth and finite, nonnegative values normalised within 10^{-12} elsewhere.

Site filter

Order (EC11). All sites are kept, with the reason recorded, when the filter is off (the default for simulations) or when there are fewer than `minimum_calibration_sites` = 1,000 SNPs. Otherwise r1, r2, r3, r4, an initial r5, the library check and the r5 recount run in that order. Removal is the logical OR of r1–r5, or of r1–r4 when `drop_uninformative_sites` = false; the default is true.

Classification (EC04). The classification uses `classification_error` = 0.01 and `classification_heterozygote_alt` = 0.5. Frequencies start at 1/3 with a floor of 10^{-6}. EM runs for at most 60 iterations, stopping from the fifth on once the gain falls below 10^{-7}. W_j, D_j^{hom}, A_j and H_j are moments from the last E-step. The leakage terms use the final frequencies:

$$\lambda_j = \sum_{i:\, d_{ij} > 0} [f_{j1} \sum_a \mathrm{Bin}(a; d_{ij}, \tfrac{1}{2})\, w(a, d_{ij}) + f_{j0}\, w(0, d_{ij}) + f_{j2}\, w(d_{ij}, d_{ij})],$$

$$\gamma_j = \sum_{i:\, d_{ij} > 0} d_{ij} [f_{j0}(w(1, d_{ij}) - w(0, d_{ij})) + f_{j2}(w(d_{ij} - 1, d_{ij}) - w(d_{ij}, d_{ij}))],$$

where $w(a, d)$ is the posterior-weighted wrong-read count of a cell.

Clean core (EC05). The histogram covers the 0.5th–99.5th percentiles with bin width $2\,\mathrm{IQR}\,n^{-1/3}$ (Freedman and Diaconis 1981). Integer data use rounded widths of at least 1; continuous data use 50–2,000 bins. A moving average over `histogram_smoothing_bins` = 5 bins smooths the counts. The mode is the first maximal bin, a reversible convention; tied modes are counted. Half-maximum distances divided by $\sqrt{2\ln 2}$ give the two standard deviations.

The local posterior is $\min(1, \hat{f}_0/\hat{f})$. Here $\hat{f}_0$ is the side-specific Gaussian at the peak density, and $\hat{f}$ is the smoothed density normalised by all finite observations, including those outside the histogram range. The window is the contiguous run of bins around the mode with posterior at least `core_posterior_level` = 0.5. It is unbounded where it reaches a histogram edge.

r1, r2 (EC06–EC07). r1 rejects only depths strictly beyond the window. r2 sums depths over the r1-passing sites before dividing, and fits its core to $\log X_j$ at the r1-passing sites with $X_j > 0$. Without such sites, r2 is not applied.

r3 (EC08). The domain is sites passing r1–r2 with $D_j^{\mathrm{hom}} \geq 1$ and $\gamma_j > 10^{-9} D_j^{\mathrm{hom}}$. The fit starts from clean weight 0.6, $\varepsilon_0 = \mathrm{clip}(\frac{1}{2}\ \mathrm{median}\,[\max(W_j - \lambda_j, 0)/\gamma_j], 10^{-5}, 0.1)$, $\varepsilon_1 = 10\varepsilon_0$ and $\phi = 1$. Each iteration updates three blocks:

- the weight, held within $[10^{-6}, 1 - 10^{-6}]$;
- $\varepsilon_0 \in [10^{-12}, 0.5]$, by safeguarded Newton steps on its concave objective;
- $(\log \varepsilon_1,\ \log \phi)$, by projected Newton ascent within $\varepsilon_1 \leq 10$ and $\phi \in [10^{-3}, 10^{4}]$.

It stops when the summed relative change falls below 10^{-5}, or after 300 iterations. The rule applies if $\varepsilon_1 \geq$ `minimum_artefact_separation` $\times \varepsilon_0 = 3\varepsilon_0$. It removes sites whose clean posterior is below `clean_posterior_level` = 0.5.

r4 (EC09). The domain is sites passing r1–r3 with $H_j \geq 1$. A component has mean m and inverse concentration t, with latent SD $\sqrt{m(1-m)t/(1+t)}$; $t = 0$ is binomial. Five fixed starts around the pooled $\sum A_j / \sum H_j$ are each fitted by L-BFGS-B, with at most three restarts and eight Newton polish steps. A start is certified when its projected gradient is at most 10^{-8}; the best start has the lowest objective among certified starts (among all, if none is), ties by parameters, then start. The clean component has weight $\geq 1/2$.

With scale $\sigma_{\mathrm{sep}} = \max\{\text{clean latent SD}, \sqrt{m_0(1-m_0)/\ \text{median}\ H}\}$, the rule applies when the artefact latent SD, or $|m_1 - m_0|$, is at least $3\sigma_{\mathrm{sep}}$. It abstains when $|m_0 - \frac{1}{2}| - |m_1 - \frac{1}{2}| \geq 3\sigma_{\mathrm{sep}}$, or when no start is certified. This contig-level gate can miss well-resolved high-depth sites.

r5 (EC10). r5 rejects a site with fewer than `minimum_expected_carriers` = 2 expected carriers (a strict inequality). After library flags, it is recomputed on unflagged samples only. r1–r4 are not recalibrated.

Library quality check (EC12–EC15, EC25)

The check runs only where the site filter applies (by default, on for VCF input), on the sites passing r1–r4.

Eligible cells. The cohort-minor allele is ALT when cohort ALT reads do not exceed REF reads. A library carries it at a cell with $\tilde{a}$ minor reads of depth d when $\Pr(Y \geq \tilde{a}) \leq \eta/(N-1)$, where $Y \sim \mathrm{Bin}(d, \varepsilon_0)$ and η = `carrier_tail_level` = 0.05. Here ε_0 is r3's clean rate, or the classification error when r3 did not fit. A cell is eligible when the library is observed there and no other library carries the minor allele; cells with $\tilde{a} = 0$ count.

Weights and partitions. Because nearby cells share reads, each eligible cell is weighted by the reciprocal of the library's eligible-cell count in its `fragment_bp` = 1,000 bp window. Sites are cut into `artefact_regions` = 16 regions of equal site count, and each library's cells into `heterozygote_depth_classes` = 4 depth-quantile classes.

Mixture components. The null mixture has, per depth class:

- binomial error components on a log grid from $0.1\varepsilon_0$ to $\alpha_{\min}$, with ratio 1.6;
- beta-binomial own-heterozygote components in both orientations, with means at r4's clean balance (or at the ends and midpoint of r4's two means when its labelling is unresolved) and latent dispersions {0, 0.0001, 0.0004, 0.0016, 0.0064, 0.0256, 0.04};
- per-region artefact shares, under which minor counts are uniform on $0..d$.

The alternative adds $c_\mathrm{f}\,\mathrm{Bin}(\tilde{a}; d, \alpha)$, with share c_f common to depth classes and regions. α takes 24 log-spaced values in $(\alpha_{\min}, \alpha_{\max}]$, where $\alpha_{\min}$ = `minimum_foreign_fraction`/2 = 0.005 and $\alpha_{\max}$ = `heterozygote_floor_ratio` $\times\, h_{\mathrm{lo}}/2$, with h_{lo} the lower end of the own-heterozygote balance interval. The mixtures are fitted by EM with at most 400 iterations and tolerance 10^{-10}. Alternatives are warm-started in ascending α, and the null is re-polished from the best alternative's weights.

Flagging. A library is flagged when $\Lambda = \max_\alpha 2[\,\ell^{(1)}(\alpha) - \ell^{(0)}\,] \geq u_N$, the difference of maximised alternative and null log-likelihoods at the first maximum. The threshold u_N satisfies $\Pr(\chi^2_1 \geq u_N) = 2\eta/(24N)$. This is a Bonferroni level under the boundary law $(\chi^2_0 + \chi^2_1)/2$ (Self and Liang 1987). It is nominal only for a regular, correctly specified likelihood and is documented as a working threshold.

Inapplicability. The check is recorded as inapplicable when any of the following holds:

- fewer than `minimum_samples` = 20 samples have observed cells;
- no cells are eligible;
- $2\varepsilon_0 \geq \alpha_{\min}$;
- $\alpha_{\max} \leq 1.25\,\alpha_{\min}$.

Effect of a flag. A flag hides the library's cells only in the founder-construction view and in r5.

Inherited segments (EC15). A diagnostic report, never a flag or parent call, groups near-private cells shared with one partner (gaps below `segment_gap_bp` = 1 Mb; at least `segment_minimum_events` = 3 events).

Discovery sites and blocks (EC24). Discovery sites have an integer total depth of at least `minimum_total_depth` = 5. Blocks are runs of `block_sites` = 200 consecutive retained sites, the last possibly shorter, and ignore spacing and the discovery mask.

Genetic map (EC27)

A map needs at least two knots, with finite, nonnegative, strictly increasing positions and finite, nondecreasing cM values. The model coordinate interpolates between knots and extends beyond them at `rate_cm_per_mb` (default 5.0). Without a map, it is that rate times position. The measured coordinate equals the model coordinate if `rate_is_measured` is true. Otherwise (the default), measured coordinates clip positions to the terminal knots, and a chromosome without a map has zero measured length. The Haldane fraction is computed as $-\frac{1}{2}\,\mathrm{expm1}\,(-2\delta)$ (ledger entry L1); each consuming model sets its own probability bounds. The TroMau configuration supplies no map file and uses the 5 cM/Mb fallback with `rate_is_measured` false.

Tie rules and shared emissions (EC30–EC33, EC35)

Shared tie quanta are powers of two, set per decision kind from the decision's scale and the contrast's measured motion under 10^{-6} relative genotype-likelihood noise. Classes link consecutive sorted exact values. A class wider than `WIDE_CLASS_QUANTA` = 4 quanta is ordered by value inside. Improvement is strict; content keys are exact discrete data. Reversal flips only arbitrary conventions, such as the histogram-mode tie, and enters the derivation. The read-model representative and r4's best start use lexicographic parameter order. Relatedness uses `gain_tolerance` = 10^{-6} nats with evidence keys.

Founder-pair emissions (EC35; assembly and founder refinement) score a sample's normalised triple P under an ordered founder pair as $P(g)$ for called dosage g, $\frac{1}{2}[P(0)+P(1)]$ or $\frac{1}{2}[P(1)+P(2)]$ for one called allele (0 or 1), $\frac{1}{4}P(0)+\frac{1}{2}P(1)+\frac{1}{4}P(2)$ for two unknown alleles of different rows, and $\frac{1}{2}[P(0)+P(2)]$ for two copies of one unknown allele (a per-observation approximation). The robust score of a predictive likelihood ϱ is $\max(\log(0.99\varrho+0.01/3),-2)-\log\frac{1}{3}$; uniform or zero-sum triples contribute nothing.

Relatedness (RL01–RL19)

Inputs (RL01–RL03, RL05). Contigs are concatenated in configuration order, fixing ranks and reduction order. Allele frequencies take `iterations` = 6 EM steps from `initial` = 0.5, with update $p \leftarrow N^{-1}\sum_i[\frac{1}{2}\omega_{i1}+\omega_{i2}]$ (ω_{ig} posterior genotype probabilities) clipped at `clip` = 10^{-4}. Every sample counts in N, and unobserved samples contribute their prior weights. Site gates are `minor_allele_frequency` = 0.05 and `minimum_observed` = 0.5, both inclusive. Thinning keeps every `family_site_step` = 20th passing site, starting from the first. Log terms are floored at 10^{-300}.

Keys (RL07). A sample's evidence rank is the dense lexicographic rank of its full child-term vector, ordered by site and then by genotype pair. Equal vectors are exchangeable. A family key lists member ranks, then parent ranks (latent = −1), each in descending order. A pair key is the two family keys, sorted. Ties within `gain_tolerance` go to the larger key.

Agglomeration (RL09). Pairs whose full-sib gain is strictly positive and strictly above their half-sib gain are merged best-first, with affected entries rescored, until none remain.

Parent assignment (RL10). Each family's first free slot is offered the best sequenced non-member (excluding members and the other parent). In families of two or more, each member i is also offered: i moves to its best other placement and fills the slot. Offers are

ranked on the sweep's starting partition by gain, parent rank and family key, and applied only if:

- its rescored gain exceeds $\log N$;
- neither its family nor its parent has been used in this sweep;
- its slot is still free;
- the parent graph stays acyclic.

No second-best search follows a failure.

Moves (RL11–RL12). A sample may not join a family whose sequenced parent is itself or its descendant. The best-improvement search applies, up to N times per sweep, the rescored largest move gain above 10^{-6} (ties by evidence rank, then target key, the singleton target included). The content-order search visits each sample once, by decreasing rank, moving it when its best gain exceeds 10^{-6}.

Merges and sweeps (RL13–RL14). Two families are compatible when their sequenced parents number at most two, none being a member of either family. Latent pairs must also beat their half-sib link. Offers above 10^{-6} are ranked by gain, then pair key. The first acyclic merge is applied, and a cyclic trial is undone without reordering. A sweep consists of parent assignment, moves and one merge, up to `max_sweeps` = 30. Convergence and the margin between the searches are recorded.

Screens (RL15). Float32 screens bound their error by $0.1 + 10^{-6}M$ nats plus $2(M+16)2^{-53}$ times the summed log-likelihood magnitudes. Comparisons the bound leaves undecided are evaluated exactly. Caches hold only gains of unchanged families; screens never change decisions.

Support (RL16–RL17). Each chromosome's measured length between family sites is cut into $\max(1, \lfloor \text{length}/1\,\text{M} \rfloor)$ equal units (`support_unit_morgans` = 1). Cuts map to the first site at or above them; empty units are dropped. A member wins a unit when its stay value, $\ell_F - \ell_{F\setminus i}$ (or ℓ_F if alone), beats both alternatives by more than 10^{-6}. The alternatives are its best join gain $\ell_{G\cup i} - \ell_G$, excluding its own family and families it parents, and being alone with latent parents. A latent one-member family therefore scores 0. Parent support compares ℓ_F with the named parent and with a latent parent; it is missing for latent slots.

Links and numbering (RL18–RL19). After the kept fit, even one stopped at the sweep cap, every pair of fully latent families with a link gain above $\log N$ is reported, with no full-sib comparison required. Families are numbered by the rank of their smallest member.

Pairwise IBD (RL04). The chain uses every `ibd_site_step` = 10th retained site, without the family-site gates. For two samples' triples P, P', $\psi = \psi_j$ and $p = p_j$, the state emissions are

$$o_0 = \Big(\sum_g \psi(g)P(g)\Big)\Big(\sum_g \psi(g)P'(g)\Big), \qquad o_2 = \sum_g \psi(g)P(g)P'(g),$$

$$o_1 = (1-p)^3 P_0 P'_0 + p^3 P_2 P'_2 + p(1-p)P_1 P'_1 + p(1-p)^2(P_0 P'_1 + P_1 P'_0) + p^2(1-p)(P_2 P'_1 + P_1 P'_2).$$

Sites not jointly observed emit 1. The chain starts uniform and stays with probability $\exp(-\theta\delta_j)$ over the model-map distance δ_j from the previous chain site, with θ = `ibd_switch_rate` = 4 per Morgan; it leaves to each other state with probability

$[1 - \exp(-\theta\delta_j)]/2$. Posteriors are averaged over the jointly observed sites. The IBD0 (IBD2) ratio compares the full chain with one lacking IBD0 (IBD2).

S2. Local founder discovery and panel selection

Local founder discovery

Inputs (DI01). Blocks hold `block_sites` = 200 sites; discovery sites have total depth of at least `minimum_total_depth` = 5 reads. Samples are sorted by name for the search and returned in input order. A block is informative when some discovery site is observed in some sample. The ceiling $\min(2N_a, 2^M)$ limits births only; there is no other cap on K.

Tie quanta (DI02). Quanta are fixed once per block from its null NLL $\mathrm{NLL}_0 = \sum_i \sum_{j\,\mathrm{obs}} [-\log \max_g p_{ij}(g) + 2\lambda]$, an upper bound on any panel's NLL. Each is the smallest power of two at least $\max(\mathrm{floor}, \mathrm{factor} \times \mathrm{NLL}_0)$: NLL comparisons 2^{-10} and 2^{-20} (q_{nll}); A-step bands 2^{-13} and 2^{-26} (q_{sample}); H-step allele changes 2^{-12}; improvement tests 2^{-10} (q_{imp}); release cells 2^{-14} (q_{cell}); probabilities 2^{-15}. Each floor is at least 16 times the measured motion of its quantity under a 10^{-6} relative perturbation of the genotype likelihoods, so no decision turns on numerical noise. Tie classes are formed by consecutive-gap single linkage; inside a class, stated preferences (such as fewer founders) precede an arbitrary content order, except that value order comes first in a class wider than four quanta. Budgets admit whole classes: a class straddling a budget is admitted up to twice the budget and omitted whole beyond it (recorded); classes ranked below a filled budget are ordinary truncations.

Fixed-K fit (DI03–DI09). λ = `unknown_copy_penalty` = 0.5 nats; τ = `switch_penalty` = 10 nats; `snps_per_bin` = 10 (one site per bin in blocks of at most ten discovery sites); likelihood floor 10^{-12}; `max_iterations` = 50. The A-step band is $[\min, \min + q_{\mathrm{sample}}]$; the category preference (row pairs, (k, U), (U, U)) is never reversed; the meet is order-free, e.g. $\{(k, k'), (k, k'')\} \mapsto (k, U)$. Samples without observed sites take (U, U). In the H step homozygous carriers count twice in usage and once in evidence; unused rows keep their alleles; a comparison that an ordinary floating-point sum cannot certify is re-summed exactly. The second usage-tie schedule runs only if the first meets a tie. A cycle returns its lowest-NLL state (ties keep first-visit order) and is never reported converged. Each start contributes its assignment-only solution and refitted endpoints; endpoints with duplicate rows are refitted recursively at lower K. Panels are canonical (rows in lexicographic order), deduplicated by content (a later fit replaces an earlier only if better by more than q_{nll}) and ordered by NLL class, smaller K, then content. Panels differing only at alleles no assigned carrier observes are one archive entry. Memoisation and fixed-point certificates (DI07) change work, not endpoints.

Cavity score (DI10). α = `state_concentration` = 0.5, ω = `unknown_prior_mass` = 0.02, δ = `double_unknown_fraction` = 0.25; base masses are $(1-\omega)/K^2$ for (k, k), $2(1-\omega)/K^2$ for (k, k'), $\omega(1-\delta)/K$ for (k, U) and $\omega\delta$ for (U, U), renormalised. A held-out sample tied among states $\mathcal{I}_i$ removes $1/|\mathcal{I}_i|$ from each. Mean-field ALT marginals μ_{kj} start from h_{kj}, $1 - h_{kj}$ and 1/2; rows without remaining support stay at 1/2; a logit within q_{nll} of zero is set to zero, keeping symmetric evidence symmetric; each start runs at most `mean_field_max_iterations` = 100 sweeps to `mean_field_tolerance` = 10^{-10}. Starts are compared by training evidence lower

bound within q_{nll}; all converged winners (all starts if none converged) form connected classes at 10^{-10}; their predictive likelihoods, never their marginals, are averaged with equal class weights before the `likelihood_floor` $= 10^{-300}$ and logarithm. No held-out evidence sets these weights. Unconverged solves are counted and block certification but remove no candidate.

$K = 1$ **starts (DI11–DI13).** Similarity is $\bar{s}_{ii'} = M^{-1} \sum_j \sum_g p_{ij}(g) p_{i'j}(g)$, distance $\max \bar{s} - \bar{s}$. Level-wise HDBSCAN* merges equal edges simultaneously and selects by excess of mass (`seed_min_cluster_size` $= 2$). A cluster's mean ALT fraction $\frac{1}{2}p(1) + p(2)$ is hardened at 1/2; within 2^{-15} of 1/2 the population ALT fraction decides. Seeds enter by whole cluster-size classes, largest first, up to `data_seed_count` $= 8$ (straddle 16); without cluster seeds, dosage rows of the most decisive samples (most observed sites, then largest summed maximum likelihood) are used. Fitted starts are kept by whole NLL classes up to `data_start_beam_width` $= 6$ (straddle 12); if every seed class is omitted the block is unresolved. Above `seed_landmark_budget` $= 256$ samples, clustering uses at most 256 landmarks lifted back to all samples (DI12); fitting and scoring still use every sample.

Ordinary moves (DI14–DI16). Deletions precede gauge rewires. Maximum cuts are exact per component up to `exact_cut_max_k` $= 12$ rows, otherwise found by degree-ordered local search (a recorded limitation); tied maximum cuts form one family with nominal budget $4 \times$ `max_cut_ties` $= 16$, admitted whole up to 32 and omitted whole above. Each cut yields a sitewise start (a site flips only if the samples with both copies on one side gain more than q_{nll}) and a start flipping all eligible sites. Deletions and rewires are admitted by whole families up to `max_proposals_per_expansion` $= 128$ (straddle 256). Data births rank samples, over both schedule endpoints of the parent, by classes (q_{sample}) of excess cost over the per-site genotype oracle, then more observed sites. Each sample supplies, for both extremal dosage families, subtraction seeds $\max(0, \min(1, \text{dosage} - h_k))$ for every row k and direct dosage rows (dosage 2, or dosage 1 or missing where the population ALT fraction exceeds 1/2, becomes ALT). Whole tie groups are collected up to $\max(16, 2(K+1)) \times$ `data_birth_attempts_factor` (3) starts; a group crossing `data_birth_tie_budget_factor` $= 2$ times that is omitted whole. Fitting stops at $(K+1) \times$ `data_birth_children_factor` (3) distinct complete children; collapsed endpoints are discarded; with no complete child, assignment-only fits are kept by NLL classes up to $K + 1$.

Soft-residual births (DI17–DI20). Rows with at least `min_soft_responsibility` $= 0.25$ expected copies in a sample (diplotype posterior with a uniform pair prior) are subtracted; the residual ALT probability, given subtracted allele h, is $P(\cdot \mid h+1)/[P(\cdot \mid h) + P(\cdot \mid h+1)]$, called at `residual_call_probability` $= 0.8$. Distances are Hamming fractions over jointly called sites (distance 1 below `min_residual_joint_known_fraction` $= 0.10$). Groups of diameter at most `max_cluster_hamming` $= 0.10$ are all HDBSCAN* tree clusters within the bound plus a greedy edge-clique cover of the remaining records, budgeted per connected component, `max_alternative_groups` $= 64$ (straddle 128). Pools are global and per (subtracted row, dominant partner) route; a record joins every route whose partner mass is within the probability band of its largest, so no convention selects one route. A consensus pools weighted log odds, one vote per sample capped at weight 1, and is emitted with at least `min_unique_sample_support` $= 2$ samples, effective support `min_effective_sample_support` $= 1.5$, a fraction

min_candidate_known_fraction = 0.8 of sites called at candidate_call_probability = 0.9, and when it is no near duplicate (dedup_hamming_fraction = 0.005 over min_dedup_joint_known_fraction = 0.6 joint sites, coverage min_dedup_bidirectional_coverage = 0.95 both ways) of a panel row or better-supported emission. Candidates enter by whole support classes up to 128 (straddle 256) and are hardened at 1/2 as for $K = 1$ seeds. Replacements delete any row of a birth, taking births by NLL classes up to max_replacement_children = 48 starts (straddle 96).

Archive, expansion and status (DI21–DI25). Each K keeps whole NLL classes up to beam_width = 6 (straddle 12); a wider leading class empties the beam while its last scored class stays selectable. Every fitted panel is remembered: one that has left the beam re-enters only with an NLL better by more than q_{nll}. Refresh visits K by $|K - K_{\mathrm{sel}}|$; new scores count against $96 + 12K_{\max}$ (exact_score_backstop_*), and a class that cannot be scored whole leaves the previous one in place. Score classes chain across K; in a class wider than four quanta only the exact maximum survives; the smallest K in the class is selected, so another K can link a class but never wins. All parents of a class are expanded before reselection. Replacements run after a phase without improvement above q_{imp}; bridges run in the residual phase when $K + 2$ is within the ceiling and $K + 1$ is scored. Expansions stop at $32 + 8K_{\max}$ (expansion_backstop_*). Stop reason, omissions, certification, ambiguity, score margin (selected score minus the best other K) and unresolved status are recorded and never steer the search.

Release (DI26–DI27). min_call_probability = 0.85, min_supporters = 2. A carrier contributes, and counts once, only when its cost difference exceeds q_{cell} in magnitude. Tied winners are matched by known content: a row absent from any alternative is withheld, and a call survives only where all matching rows agree, with the least confident probability and smallest support.

Cost. An A step costs $O(NK^2B)$; a trajectory iteration $O(NMK^2)$ (at most 50 iterations, two schedules); the search $O(NMK^4)$ under exemption 0c ($K^4 = 810,000$ at $K = 30$); a mean-field sweep $O(NMK^2)$; release $O(NKM)$. Budgets are scientific parameters, independent of computing resources.

Local panel selection

Model (LS01–LS02). u = unknown_mass = 0.01; γ = redraw_rate_per_morgan = 3, a redraw intensity rather than a crossover rate. Recursions are scaled and samples are summed in ascending order, so results do not depend on threads.

Objective (LS03–LS04). Context founders without calls are dropped; founders with equal rows remain separate claimers. Among descriptions within a relative 10^{-9} of the shortest, the largest matching is taken. Edit distances count called sites only, so one changed call moves the code by at most one edit increment.

Fit (LS05–LS07). tolerance = 10^{-6} nats. Rows whose summed gains lie within a relative 2^{-10} (FLIP_TIE) of the largest are compared by the exact objective after flipping; if different panels remain tied the fit stops. Updates stop at $20 + 4K + 2M$ (update_backstop_*); rows are returned in canonical order. The trajectory-best state is the earliest strictly best; its frequencies are then refitted while the context objective rises. A snap proposes each called site where a matched row differs from its founder, unless the flip would duplicate a

row, with bound equal to the exact one-bit gain plus the edit saving $\log V(n,d) - \log V(n,d-1)$. It needs a bound above η; bounds within η of the largest form one class, decided by the smallest canonical panel; a snap is kept only if the objective rises.

Search (LS08–LS15). Deletions are proposed in row order, merges in lexicographic pair order and additions in bank order (new row frequency $1/(K+1)$); a single-row panel has no deletions or merges. A merge's consensus row takes the smaller row's allele within 2^{-20} of 1/2; the merged row carries $f_k + f_{k'}$, and a consensus equal to another row merges that row too. Starts with equal panels and frequencies within 10^{-12} are one start. The merge bound is the deletion panel's log-likelihood plus the kept row's expected gains at changed sites plus $\sum_k \bar{n}_k \log(f_k^{\mathrm{m}}/f_k^{\mathrm{d}})$ ($\bar{n}_k$ and f^{d} from the deletion panel, f^{m} the merged frequencies on its rows), the larger of two such bounds. Exactly scored merges are, per row, the best-bounded merge with a nearest row and the best-bounded merge not involving it whose consensus is nearest it. Each kind refits whole classes until `refits_per_kind` = 8; a class crossing `refits_per_kind_cap` = 16 is excluded. Refits reaching one panel are one candidate. Rounds stop at $10 + 2K$ (`round_backstop_*`). If a start's fit reaches the previous round's panel, the previous fit is kept unless beaten by more than η; if the previous fit is better by more than η, the search starts from it. A whole search is reused only for identical inputs; fits, scores and bounds are memoised across the first two rounds (LS13, changing work only). Outcomes of several starts are compared by objective class, then fewer founders; distinct tied panels become alternatives (LS14). A retained incumbent discards the attempted alternatives (LS15).

Bank (LS16–LS20). The initial round has no incumbent and no bank search. A proposal row with no call is dropped; a partial row takes, at uncalled sites, the alleles of every latent row at minimum Hamming distance; tied completions give one source per combination up to `completion_variant_cap` = 8, above which the source keeps only uniquely completed rows. A source's best set is the nested class of cavity score, then NLL, then fewest founders; each member is released by discovery's rule, with uncalled cells taken from the source's own latent panel. The bank search first fits the whole bank as one start, then visits deletions, additions and row replacements (closest first) with κ = `complexity_scale` = 0.5, reselecting by nested classes of penalised score, NLL and founder count, and stopping without improvement above q_{imp} or after $8 + \lceil K/2 \rceil$ rounds. It fits at most $512 + 64K$ novel starts per call and admits at most $1024 + 128K$ panels, overlapping fit families entering by whole NLL classes (LS20). The search bank is the latent rows, starts, selected panels' rows and the previous panel's rows.

Release (LS21–LS24). `min_call_probability` = 0.85, `min_supporters` = 2, `favourable_margin` = 2^{-14} nats. Carrier probability counts a homozygous state once; a carrier whose own gain lies within the margin of zero is not favourable. A previous release is reused only for identical panel and frequencies without context witnesses. A site's unknown-copy count is the smallest count whose mass is within a relative `unknown_count_band` = 2^{-15} of the largest; samples without depth report 2; tied alternatives take the maximum. Rows shared by all alternatives keep agreed calls, mean ALT probability and smallest support.

Context refit (LS25–LS28). `context_levels` = (1, 2). An uncovered block receives its current calls without context founders, hence no witnesses, claims, trajectory-best adoption or snaps. A component must match its blocks' concatenated positions exactly; duplicate component rows give a consensus template. Bins are blocks, spaced by the map

interval between block centres; founders have prior mass $(1-u)/K$ and the background state u. Uncalled founders and the background carry the site's ALT frequency $(1+n_{\text{ALT}})/(2+n_{\text{called}})$ over called founders; the emission mixes the expected likelihood with $1/3$ at weight `emission_mix` = 0.01. A site's founder-allele posterior, under a uniform prior, is enumerated exactly for at most 10 founders; above that, a belief-propagation mode and local ascent fix certain founders (flips costing over 20 nats), withhold multimodal sites, and enumerate the remaining core exactly (at most 10 founders) or within a Hamming ball of at most 1,024 configurations (radius at least 3), with a shell guard. Calls are released at `posterior_threshold` = 0.99 with count at least `min_support` = 2 or mass above $2+2^{-20}$; other cells keep the component's call, then the template. Duplicate local rows sum support and record multiplicity. Recorded summaries do not affect decisions (LS29–LS31).

Cost. A forward–backward pass is $O(NMK^2)$; a fit at its backstop $O(NMK^2(K+M))$; a climb $O(NMK^3(K+M))$ under exemption 0c; context-coded screening $O(K^5)$ per round under exemption 0 (K^2 merges each with an $O(K^3)$ matching; $24,300,000$ at $K=30$); snaps $O((K+M)^2)$ passes; context refit $O(NBS^2)$ time and memory per component.

S3. Assembly, founder release and refinement

Parenthesised identifiers are the software's method units; defaults are those of release 4.4.0 (none is a user setting). Notation: T bins of a batch; W units of a proposal window; b_w surrogate bins of a window; R the window's row-alphabet size.

Founder completion

Folds (U03). `fold_seed` (1729) is hashed with each sample's observation-mask content and depth key; equal-key groups, visited in byte order, go whole to the lighter fold (ties and inactive groups to fold zero). No random draw or label chooses a fold.

Cross-fitted variational EM (U04). Samples copy unordered row pairs and unresolved alleles are binary variables shared across samples, both under uniform priors; emissions are $\max\{\log(0.99\bar{p}+0.01/3), -2\}$. The variational family factorises over samples (pair states) and sites (configurations). A site's support is the full cube up to `MAX_EXACT_BITS` (10) unresolved founders. Above that it is an uncertain core fixed from the first iteration, so the ELBO stays a lower bound. One start runs 2–100 iterations (ELBO tolerances 10^{-8}; a decrease beyond 10^{-8} is an error). Training-fold calls need posterior ≥0.99, ≥2 carriers, non-exchangeable founders, a unimodal site and a passing shell guard. The held-out fold is painted only where the released panel calls every founder; absent training data leaves the pre-fill panel (recorded).

Cavity fills (U05). Bins hold 50 SNPs (`snps_per_bin`), and outside-bin statistics are prefix plus suffix sums. A founder is ineligible if another founder has an identical outside-bin allele vector (tolerance 10^{-12}) or differs from some founder at fewer than one observable outside-bin site. Support, over focal-informative samples sensitive to the target, is the larger of the number with leave-bin-out carrier probability above 1/2 and the sum of those probabilities. A fill needs support ≥ 2 and posterior ≥ 0.99. A single-bin multi-founder block therefore gets no fill.

Site posteriors (U08–U10). Dosage polynomials are enumerated exactly; a value cancelling below 10^{-10} is recomputed as a direct positive sum. Above ten unknown founders, damped loopy belief propagation (damping 0.5, tolerance 10^{-9}, ≤500 iterations) (Murphy, Weiss, and Jordan 1999) is followed by deterministic ascent from its mode and complement (backstops 50), on an uncertain-core cube or a Hamming ball (≤1,024 configurations, radius ≥3, 3-nat shell guard). Non-converged BP is flagged; this can withhold a fill even when a mode exists.

Boundary links (U06). Rows with identical pre-fill vectors form classes. Two classes across a boundary are linked only if, on samples informative on both sides, the mean squared discrepancy of expected copies and of $\min(\text{copies}, 1)$ is ≤0.099; the pair is a reciprocal unique minimum by margins 0.03 (pooled) and 0.01 (each fold); there are ≥6 pooled and ≥3 per-fold informative samples; each class has ≥1.0 pooled and ≥0.5 per-fold effective copies and one or more anchor sites separating it from the other classes of its block; and both blocks have two or more classes. Exact best/runner-up ties are not broken by row number.

Occupancy gate (U07). A block releases nothing if selection left unexplained copy mass above `maximum_discovery_unknown_mass` (0), or if it has a single founder.

Abundant route: squared carrier probabilities sum to ≥8 and ≥5 samples carry the row with probability ≥0.8; all its candidates are released (a sum within the tie band of 8 is not abundant). *Rare-continuity route:* through an accepted single-row link, ≥2 samples informative on both sides carry the row (≥0.9) and the neighbour (≥0.8), product ≥0.72; the best neighbour must be unique (tied neighbours abstain only when their qualifying samples differ); fills are released only where a qualifying sample is focal-informative.

Released fills publish the allele with posterior above 0.5.

Hierarchy

Partition (U11). Linking proxies stride to at most `max_sites` (2,000) usable sites per component. A segment of n components is cut into $m = \lceil n/10 \rceil$ batches, plus one when n is odd and m even. Sizes are near-equal, with the larger ones in mirrored pairs. Breaks persist. A batch with one component or no common informative sample passes through unchanged.

Levels (U02, U12). Stop reasons, checked on unrefined join output, are in order: single component; no reduction (level ≥2); only phase breaks (level ≥ `minimum_levels`, 4). Joined founders are sorted lexicographically, so later slot tie-breaks are content rules. Final-role refinement follows every level, with additions only at the level whose stop reason is set. First-level paths, refined first and then unrefined, form the coarse alphabets of later refinements. Local selection runs one or two unrefined levels as context.

Alternatives between levels (U13). A family is an input choice only if it covers a component's exact span and contains that component's nominal panel; otherwise it stays unresolved. If the product of choices is at most `BRANCH_LIMIT` (16), every joint branch is joined on the sites usable under all choices and one common cohort; each count's winners

are its top LL tie class, and the representative has the smaller K, then fewest exact switches, then content order. A larger product is recorded as an unresolved joint budget; a tied family is never reduced to a convenient subset.

Linker and beam

Scans and EM (AJ05–AJ07). Emissions are $0.99\,\Pr(\text{obs} \mid \text{class}) + 0.01/3$ over seven predictive pair classes, without a floor. Inside a component a homologue switches with θ equal to the interval's map distance in Morgans (bp times the fallback rate without a map), capped at 0.5; weights $(1-\theta)^2$ (stay) and $\theta(1-\theta)/(K-1)$ per other founder are normalised. An error-tract chain has stationary fraction 0.01 and mean tract 20 kb; error-state emissions are the mean of the three likelihoods. EM per gap starts uniform; the M-step adds 0.1 pseudocounts, floors logs at -10, renormalises and mixes 0.01 uniform; it stops at $\max |M(\mathbf{A}) - \mathbf{A}| \le 10^{-3}$ or after 50 full updates (recorded as unconverged).

Horizon and beam (AJ08–AJ11). The horizon is $G = \max\{1, 1 + \lfloor 0.5/(3\bar{D}) \rfloor\}$ (`recombination_tolerance`, `gap_horizon_scale`). The beam builds paths forward with 16 per endpoint row, then refines block by block backwards, and archives every selected path. With constant transitions (AJ10, retained accretion), only $O(BK)$ first-covering paths are proposed. Linkage (summed step scores) is consulted only within LL tie classes.

Join panel search

Objective (AJ01–AJ04). The bin size is $\max\{\min(100, \max(10, \lfloor \bar{L}/20 \rfloor)), \lceil L_{\text{span}}/2000 \rceil\}$, with $\bar{L}$ the mean number of sites per input. The switch source is the first maximiser in pair order, and a state stays on equality. The exact minimum switch count is only a secondary prior. Proposals come from tracebacks under both conventions.

Start and tier 1 (AJ12, AJ15, AJ16). All anchor panels are scored. Proposal kinds run in order: replace, static replace, novel replace, local, one-sided, splice, cycle, switch-splice, switch-cycle. An exploratory painting at penalty $\min(\lambda, 10)$ exposes coordinated switches for proposals only. At most `full_scores_per_kind` (16) edits per kind are scored exactly (a cut class kept whole, up to twice that). A move improves when $2\Delta\text{LL} > \max\{10^{-8}, 10^{-9}(1 + 2|\text{LL}|)\}$.

Admissibility (AJ14). A move must keep K distinct paths, and each changed path must keep a private cell and stay more than `near_identity` (0.01) from every other path. Admissibility is checked on the edit witness, before deduplication; a seed and its copy are exempt during their refit.

Bounds and tier 2 (AJ17–AJ20). Exclusion forwards, the maxima without up to two paths, are recomputed only where those paths cover a bin's maximiser set; this is exact under ties. Two-region bounds limit removals of up to two paths with additions of up to two. Only a certifying bound skips work; samples whose bound exceeds their base by $10^{-9}(1 + |\cdot|)$ are scored exactly. Tier 2 enumerates pool replacements and the local, one-sided and splice moves in canonical order, scores them best-first and records flat neighbours.

Pool, plateaus, replays (AJ26, AJ28, AJ29). Quotas scale with row multiplicities. The quota ladder doubles to 256, and to 1,024 after an acceptance at 256. A pool with identical ordered content triggers no count work. Plateau continuation follows uncertified

winning-family members. Up to MAX_BRANCHES (4) searches per join, the primary included, replay forced choice prefixes (FIFO); a mismatch or exhausted budget records incomplete resolution, and seed choices are never forced.

Backstops (AJ25, retained accretion). With K_b the founders of input b, a search stops after $100 + 20 \max_b K_b$ sweeps. Births and seed swaps freeze after $50 + 10 \max_b K_b$ count changes, while deaths continue. Backstop stops are reported as uncertified; a final census of paths failing their own death test is reported only.

Join count step

Deaths and births (AJ21, AJ22). Paths without novelty are deleted one by one, starting with the largest LL($\mathscr{P}f$). Otherwise the best candidate is ranked by that value in wide tie classes, then by negative linkage, then by content, and is tested. A birth is scored only if it passes two necessary conditions on upper bounds $u_i \geq d_i \geq q_i$: the second-largest u_i exceeds $c/(2\tilde{N})$, and $\sum u - \max u > c/(2\tilde{N})$. Survivors are scored exactly in decreasing bound order, until no bound can come within four quanta of the best accepted gain.

Duplicate seeds (AJ23). When ordinary births fail but a pure-mosaic pool path supplies a chimera indicator, every selected founder is duplicated in turn. Every seed is refitted by certified moves restricted to it and its copy, then settled (paths without novelty deleted, then a death test). A refit competes if it survives, is not smaller and raises LL; acceptance is a birth or a seed swap at unchanged count. Endpoint LL, its tie class and content order decide among survivors. Indicator and coverer ranks do not restrict seed admission.

Cycles (AJ24). Born-path tags follow same-count edits; deleting a tagged path freezes births, and is marked undecided if ambiguous in a cycle that began with a decided deletion. Revisiting a canonical count state freezes births and seed swaps.

Bootstrap and shadows (AJ30–AJ33). Bootstraps draw 2,000 resamples (level 0.95) from a stream keyed by content (decision kind and larger panel in joins; component end positions, size, round and kind in refinement), shared across a shadow family. Bands are resolved only for reported decisions. A complete shadow family of up to 2,048 members is enumerated only if the total test passes. In joins, an unverified count remains a diagnostic.

Proofs and bound tests

Count bounds (AJ34, RC10). Optimistic scores let each block choose a free panel, resetting between blocks. A block with $K + 1$ rows takes its best one-row deletion. Deletion bounds need every founder to have a private cell. With $\underline{d}_i = F_i(\mathscr{P}_K) - U_i$ (U_i the per-sample bound), a deletion round is skipped when $\sum \underline{d}/c - \frac{1}{2}$ and $(\sum \underline{d} - \max \underline{d})/c - (\tilde{N} - 1)/(2\tilde{N})$ exceed the rounding tolerance and the replicate-wise percentile of $\sum \underline{d}/c - \frac{1}{2}$ is positive.

An addition round is skipped if the total bound gain $U - F$ is below $c/(2\tilde{N})$, or if every $u_i = U_i - F_i$ is (which also proves non-ambiguity). Skips are recorded as proven decisions.

Shadow-envelope certificate (AJ35; retained accretion). In refinement only, a proof is attempted for an over-budget family when the largest per-block product is ≤2,048 and local work ≤4 times ordinary work. Block messages maximised over local products give

U_i^{σ} bounding every shadow, so $q_i \geq F_i(\mathscr{F}^{\dagger}) - U_i^{\sigma}$. The proof succeeds only if every combined field equals the incumbent's under every permutation of each bootstrap draw. A successful proof clears the count-unverified flag only.

Refinement: objective and fixed-K search

Workspace (RC01, RC02). Objective sites are kept sites with a called prepared row; at an all-uncalled site, pairs would differ only through the allele prior. Samples without evidence contribute exactly zero and are excluded from $\tilde{N}$ (F still sums all samples). Interval hazards are `reset_rate` (1 per Morgan) times map distance, floored at $-\log(1-10^{-15})$ and summed over skipped sites; the first informative site has no transition; costs are stored in single precision. A map is required; genotype error is fixed at 0.01. The recursion over unordered pairs is

$$\mu'\{a,b\} = \max(\mu\{a,b\},\ \max(\bar{\mu}_a, \bar{\mu}_b) - \kappa,\ \max\mu - 2\kappa) + \epsilon(a,b), \qquad \bar{\mu}_x = \max_y \mu\{x,y\}.$$

Exact ties prefer lower switch cost; a separate recursion counts switches (RS02).

Surrogate and windows (RC03, RS04). The bin size is $\max\{\text{floor}, \lceil L_c/2000 \rceil\}$. The floor is $\max\{200, \lceil L_{\text{chrom}}/2000 \rceil\}$ at levels 1–2 and 200 thereafter (`proposal_min_sites_per_bin`, `proposal_max_bins`). The Potts penalty is 20 nats. Fine windows of 8 blocks at stride 4 offer all prepared rows. Coarse windows of 4 first-level spans at stride 2 offer the distinct first-level paths.

Window programme (RS05, RS06; exception 0e). A box bound certifies already-optimal windows. Otherwise an exact frontier programme prunes by closed-message dominance (10^{-6}) and valid bounds. Past a frontier of 256 it restarts with Lagrangian suffix bounds (≤100 dual sweeps) (Rush and Collins 2012) and a feasibility beam of 16, keeping ≤512 children per unit; it is approximate where that cap binds, and such units are counted. A frontier beyond `frontier_bytes` (128 GiB) keeps the incumbent. The canonical answer is the incumbent when it lies within $2^{-14} + 10^{-9}|H|$ of the optimum (H the incumbent's surrogate value), and otherwise the lexicographically first path within that band.

Work account (RS07, RC18; accretion, exception 0e). Each template is allowed $2^{25} NWb_w^2 \max(K,R)^3$ charged units (`window_work_coefficient`). Exhaustion discards the template and stops the component, which keeps its input rows and families plus an unperformed-search family; during polish it keeps the unpolished panel. The bound is per template, not per component.

Acceptance (RS08–RS10). At each junction, exchanges come first, at most 4 (`EXCHANGE_ROUNDS`). Content moves follow, fully re-ranked through 4 acceptances (`CONTENT_FULL_ROUNDS`) and then rescored singly. The threshold is TOL = 2^{-9} nats. A coarse move with $\Delta F + \Delta(\text{switch cost}) < -10^{-6}$ is guarded: affected paths are traced with and without it, switch events are moved to the leftmost site of their evidence interval, paired excursions are excluded, and the move needs $\Delta F - C_{\text{shared}} > \text{TOL}$ with $C_{\text{shared}} = \sum_i \min(\text{saving}_i, \text{cost of shared one-way removed events})$. Fine moves and bridges bypass the guard.

Fits (RS11, RS12). A fit converges at a sweep that accepts nothing. Reaching `sweep_limit` (100) records nonconvergence. An F that deviates from its prediction by more than $10^{-9}\ \max(1, |F|) + 10^{-6} \times$ (accepted moves) is an error. Of the two direction starts, the winner is the higher F class, then the minimum switch count; tied variants are retained, and rows are canonicalised.

Refinement: count pass

Rounds and candidate refits (RC09–RC11). Refit the incoming panel and run ordinary deletion rounds. Candidates are all founders without a private cell when any exist, otherwise all founders; every eligible candidate is refitted. At an addition-enabled level every founder is a duplicate seed, and every seed is refitted. Optimistic bounds may certify an entire round unneeded. Compare converged endpoint F with the incumbent's scale, choose the content-smallest origin in its best tie class, then evaluate that class's verdicts. Deletion stops on the first rejection; additions stop on rejection or at twice the incoming K. Every refinement deletion uses the ordinary count rule.

Symmetric null and ties (RC12, RC13). The death class is the top tie class of unrefitted $F(\mathscr{F}^{\dagger} \setminus j)$, taken over founders without a private cell if any exist. Every member is evaluated with one bootstrap stream; disagreement or an unresolved member family yields $\mathscr{F}$ with ambiguity. Every member of the selected refit's tie class is also decided. Opposed verdicts keep the smaller-panel outcomes, which are ranked by margin, then by fewest switches.

Restoration, terminal family, continuation (RC15–RC17). A fit is remembered at its K only if it improves by more than $\max(\text{TOL}, \text{quantum})$. At the final K a decisively better fit is restored and its adjacent-K comparisons rerun. The returned panel's F class is rescored exactly; its fewest-switch member represents it and the live family is released. Continuations replay alternative input roots and count choices breadth-first from the original input, within `alternative_branches` (16) leaves. Winning classes are compared per count, preferring the smallest K, then the fewest switches. Overflow stays unresolved.

Polish (RC19, RS14–RS17). Polish applies one-direction fits to the decided panel and to each distinct live alternative. At each ordinary fixed point it tries a beam escape (width 64); if nothing is accepted, a long repair (one-founder moves over 64-block windows at stride 16, beam 16, each with an optional 8-block private correction from the two most-agreeing donors or the incumbent); if still unchanged, a bridge (fine precursor plus suffix exchange). Accepted gains are verified exactly; counts are not re-evaluated.

Ties

Each quantum is the smallest power of two that is at least $\max(\text{floor}, \text{factor} \times |\text{scale}|)$:

Kind	Floor	Factor
`join_ll`	2^{-19}	2^{-34}
`join_ll_wide` (deaths)	2^{-19}	2^{-30}
`join_proxy` (bounds)	2^{-19}	2^{-24}
`beam`	2^{-18}	2^{-30}
`refine_F`	2^{-9}	2^{-26}

Kind	Floor	Factor
`count_margin`	2^{-26}	0

A class chains values whose consecutive gaps are at most one quantum, and a class wider than 4 quanta keeps value order (U14, AJ27, RS09). Within a class, linkage decides first (joins only), then content. Convention reversal changes only arbitrary content orders, never a prior.

Founder release (U16)

Alleles that no painted carrier observes are withheld first. For each retained family, blocks whose row multisets differ are isolated and their unmatched rows withheld; greedy forward and reverse scans cut continuity wherever the panels stop agreeing up to a row permutation. Segments are reordered by content, and withheld rows are barred from the downstream r5 fill. Count-unverified records annotate segments without withholding. A family not matching the final panel is reported as unresolved correspondence.

Cost class in K

Mechanism	Dominant cost	Class
Completion	$O(NLK^2)$ cross-fit; $O(NK^3)$ ball sites; 2^{10} cube	$\le K^3$
Linker	$O(NJK^2)$ scans (J emission sites); $O(NK^3)$ propagation	$\le K^3$
Join search	Adaptive structural screen has a cubic analytic bound; direct query count has a quartic trend on a finite grid	cost register 0e (AJ17)
Count refits and shadows	$O(K)$ fits, $O(NLK^3)$ per round; repeated deletion rounds $O(NLK^4)$ worst case; ≤2,048 shadow scores per member	Complete eligible candidate domain; other search factors separate
Refinement search	$O(NLK^2)$ messages; $O(BNK^3)$ exchanges; window fixing bounded by the work account	exception 0e (RS05, RS07)
Release	$O(Bn_{\text{alt}}K \log K)$, n_{alt} alternatives	$\le K^3$

S4. Painting, pedigree inference and founder refit

Painting

Parameter names follow the generated reference; values are 4.4.0 defaults; method units are in parentheses.

Alphabet, bins and local keys. Classes are exact call signatures, numbered by first appearance; unanchored classes and BACKGROUND share one public UNKNOWN identifier (PR01). Bins split the L_s selected sites into $\max(1, \lfloor L_s/\texttt{snps_per_bin} \rfloor)$ runs, with `snps_per_bin` = 100; the first $L_s \bmod B$ runs are one site longer, independently of task windows or sample batches (PR02). In each bin, anchored classes called there with

identical $\{-1, 0, 1\}$ vectors share their smallest index as a key; other anchored classes key to themselves, and the rest to UNKNOWN. Local equality does not imply chromosome-wide row identity (PR03).

Emissions and transitions. A site is informative when it is observed and its likelihood row spread exceeds `INFORMATIVE_SPREAD` = $16\varepsilon_{\text{mach}}$. Sums run in site order and are divided once by the design effect. A bin split by label continuity keeps its whole-bin denominator in every part (PR04). `MINIMUM_SWITCH_PROBABILITY` = 10^{-15}. The switch probability θ_b never exceeds $(S-1)/S$, and edges of zero length are left out of the ϱ update. The rate ϱ is a pooled copying rate, not a count of meioses (PR05). Track states are the S faithful states followed by one error state per anchored class, $S' \le 2K+1$; the first bin draws labels uniformly and fidelity from the stationary law. Painting reads every sample, including those QC excludes from founder construction. The diploid transition factorises over tracks, so recursions cost $O(S'^2)$ per bin and sample (PR06). Forward–backward is scaled at every step and reduced in sample order (PR07).

Estimator. Starting values are `initial_switch_rate` = 2.0, `initial_error_entry` = 0.001 and `initial_error_exit` = 0.25, with `estimation_iterations` = 8 updates (PR08). With n_{tr} eligible tracks and expected label changes $\bar{n}_b \le n_{\text{tr}}$, the update for ϱ maximises

$$Q(\varrho) = \sum_{b:\, d_b > 0} [\bar{n}_b \log \theta_b(\varrho) + (n_{\text{tr}} - \bar{n}_b) \log (1 - \theta_b(\varrho))]$$

over `RATE_BOUNDS` = (0.01, 1000) per Morgan. The search takes `RATE_GRID_POINTS` = 241 log-spaced points, then `GOLDEN_STEPS` = 60 golden-section steps. The previous value is kept unless Q improves; Q can be multimodal, so the search is not a certificate. Let $n_{\uparrow}$ and $n_{\downarrow}$ be the expected error-run starts and ends, k_f and k_e the kept faithful and kept error transitions, and ι_e and ι_f the stationary draws into error and faithful states. The update for (α, β) maximises

$$(n_{\uparrow} + \iota_e) \log \alpha + k_f \log (1-\alpha) + (n_{\downarrow} + \iota_f) \log \beta + k_e \log (1-\beta) - (\iota_e + \iota_f) \log (\alpha + \beta)$$

over `ENTRY_BOUNDS` = $(10^{-6}, 0.5)$ and `EXIT_BOUNDS` = $(10^{-3}, 0.999)$. It uses L-BFGS-B in log space from the previous values and from stay-only ratios, keeping the previous values unless improved. Results are clipped and rounded to a natural-log grid of `estimate_grid` = 2^{-6}. Without an anchored class there is no error state, and (α, β) are reported at their bounds. A fitted $\alpha = 10^{-6}$ is a boundary fit, not a test of absence.

Decoding. Predecessors within `VITERBI_TIE` = 2^{-15} nats of the best are tied. Ties prefer staying in the same state, then the same label in its other fidelity state, then a label change from the smallest index (the largest under reversal); the final state is the smallest index within the quantum (PR09). At ties between ordered and swapped continuations, the suffix is swapped if the swap gives strictly fewer allele-visible changes. A change between classes that share a local key is not allele-visible, and a pure swap is no switch (PR10). Switch intervals use a discrete test on calls, the observed mask and the shared-allele rule, and stop at neighbouring switches. The two conventions' masks are combined into the smaller unordered cover. Equal-size covers that differ give every candidate to both tracks. Canonical track order is fixed right to left by fewer changes, then fewer allele-visible changes, then the smaller mask on track 0. `MAXIMUM_MASK_STATES` = 63 requires $S \le 63$; at $K = 30$, $S \le 31$ (PR11).

Release. Only q^F and q^{AF} are computed, each the minimum over the two conventions (PR12). The threshold `minimum_release_posterior` = 0.9 is inclusive, with no tie band. Statuses take precedence in this order: ineligible, low posterior, switch interval, local equivalence, then the class status. Member sets are the class's rows (singleton or pooled), the rows of all key-sharing classes (local equivalence), or the union of the candidates' rows, extended to key-sharing classes in stage-2 bins and empty if any candidate is BACKGROUND or unanchored (switch interval). A switch interval cannot release a bin that failed the staged rule, and ineligible samples report zero switches (PR13); caching affects resources only (PR14).

Label continuity. Link points are the first selected sites of atomic spans. Only singleton anchored classes move, and the search needs two movable classes, a link and two bins (PR15). Exact transposition gains separate the unaffected states from the $4S - 4$ affected ones, using left and right max-product messages, and are summed in sample order (PR16). The unit Δ_{sw} is the median over edges of the log-probability of the ordered pair staying minus that of one specified single-track change. Candidates gaining at least `label_continuity_minimum_switch_gain` = 2 units of Δ_{sw} are ranked in descending tie classes. The `nll` tie quantum is the smallest power of two at least $\max(2^{-10}, 2^{-20} \times \text{largest gain})$ nats. A class wider than `WIDE_CLASS_QUANTA` = 4 quanta keeps value order; otherwise link and pair index decide. A placement moves left while the two rows carry identical founder-axis data between consecutive links.

Acceptance requires the painted-switch total to fall by at least 2, with losers at most `label_continuity_maximum_loser_ratio` = 0.1 times winners. A tie class is skipped when another acceptable member would relabel differently, and a candidate that changes a class's error capability is rejected. Each accepted move lowers the total, ensuring termination (PR17). Verification requires the relabelled panel to have identical classes, sites and anchoring and a strictly lower repainted total; otherwise the assembled panel is kept, with a recorded reason (PR18). The stage order is model, continuity, decode, release (PR19).

Cost. The chain costs $O(NBS'^2)$, masks at worst $O(NBS^3)$, and the screen $O(\text{links}\, NS^3)$ per iteration. Label-continuity checks have an $O(K^4)$ worst case, carried as documented exception 0e (PR17). Tie peers are grouped once per iteration; each tested candidate can still require a full decode.

Pedigree scoring

Inputs. A candidate is available in a component when it has an observed site with likelihood spread above `INFORMATIVE_GL_SPREAD` = 10^{-12}. An unavailable parent reduces M1 to M0, and M2 to the other parent's raw M1 (PS01). At most `markers_per_bin` = 10 markers are kept per painting bin: all of them, or $\text{first} + \lfloor s\, n/10 \rfloor$ for $s = 0, \ldots, 9$. A component is admitted when some selected likelihood triple is non-uniform, and anchoring precedes subsampling (PS02). An information group closes once its running count of selected markers reaches `markers_per_information_block` = 100, with each empty bin counted as one. Then $\eta = \min(3, n)/n$, from `effective_markers_per_information_block` = 3, and $\eta = 0$ without informative markers (PS03). The selector probability r_b uses model distances,

fallback tails included, clipped to $[10^{-15}, 0.5]$ (`SELECTOR_SWITCH_FLOOR`). Measured lengths are clipped to the map's knots unless `rate_is_measured` (default false), and are zero without a map (PS04).

Projection. Within the source chain only the initial law is symmetrised. Evidence-free samples take a uniform initial law and unit right weights, and a loss of supported mass is an invariant error (PS05). The projection then symmetrises each bin's marginal. BACKGROUND keeps its unconditioned ALT probability when the conditioning denominator is not positive (PS06). Each bridge is $\mathrm{diag}\,(\mathrm{stay})$ plus a rank-one off-diagonal term and at most one pivot arm. Its maximum-entropy root is found by at most `bridge_max_iterations` = 128 bisection halvings. A post-check (`bridge_postcheck_tolerance` = 2×10^{-13}) verifies it; failure stops the stage, with no fallback (PS07). The excursion weight is $v_b = r_b\,(1 - \min\{1, n/\texttt{source_gap_support_sites}\})$. Here `source_gap_support_sites` = 1 and n is the smaller adjacent-bin informative-marker count (PS08). `null_prior_samples` = 1. A rounding-sized negative null persistence saturates to the uniform kernel; a genuinely negative one raises an error (PS09).

Child forward. `mismatch_probability` = 0.01. Underflowing bin products are recomputed in log space. Tempering is $\mathrm{pow}(\mathcal{E}/\max \mathcal{E}, \eta)$ for a bin emission $\mathcal{E}$; the power is skipped at $\eta = 1$, and $\eta = 0$ still normalises. The composite weight $\frac{1}{2}$ is fixed (PS10).

Certified M1 and the panel. Block-certificate bounds keep called alleles exact, relax uncertain ones to endpoint likelihoods and take the smaller of two block partitions, with outward padding (PS11). Exact queries cover, in order, the tie class of maximal bounds, the best frontier, the floor frontier and every cell able to reach a local maximum. With chromosome temperatures $t_z = \max\,(\lceil M_z/100 \rceil, 1)^{1/2}$ (`information_tempering_power` = 0.5), let $t(M)$ be the same function of the genome total of selected markers, M. A candidate is retained when its genome M1 reaches $\chi_c = \max \mathrm{M1} + \texttt{LOG_FLOOR} \cdot t(M)$, with `LOG_FLOOR` = −60. All others receive

$$\ell_{cz}^{\mathrm{floor}} = \hat{\ell}_{cz} - \frac{t_z}{\sum_{z'} t_{z'}} \Big(\sum_{z'} \hat{\ell}_{cz'} - \chi_c \Big),$$

where $\hat{\ell}_{cz}$ is the certified chromosome-local maximum. The floor is applied exactly once, because it is not idempotent (PS12). Screen utilities decide nothing (PS13).

The pair bound gives every state with $\pi \geq$ `pair_bound_high_mass` = 0.01 its own group, gives BACKGROUND its own group, and puts all remaining named states in one low group. Emission envelopes use the corners of the transmitted-ALT box, since the site likelihood is bilinear (PS14). A pair enters the panel if, in the full fit or any LOCO fit w, $w \cdot \mathrm{bound} \geq \max\,(w \cdot \mathrm{M0},\ \max_q\ w \cdot \mathrm{M1}_q) - \texttt{pair_relevance_margin}$, q retained, with margin 150, or if the gap is NaN. There is no cap on parents or pairs (PS15). In a component where only one parent of a pair is available, the pair scores as that parent's M1 (PS16).

Exposure and junctions. Exposed map length integrates the smaller of the two adjacent bins' exposures over capped measured intervals (PS17). Junction counts come from an exact minimum-switch dynamic programme over unordered pairs of deduplicated trajectories, with costs 0, 1 and 2. Unknown labels are compatible with every trajectory, and callable bins count non-negative labels on both tracks (PS18). The schedule traces

the first minimum path, or the last under reversal. Detection is $\min(1, \text{callable}/2B)$. Counts above the child's total are dropped, and zero retained mass gives $-\infty$. Equally parsimonious alternative placements are not integrated (PS19).

Exclusion e-value. At a jointly observed site j, with normalised likelihood triples ξ and ζ, define $\Psi_j = \max\{\xi_0 \max(\zeta_0, \zeta_1), \xi_1 \max_g \zeta_g, \xi_2 \max(\zeta_1, \zeta_2)\}$ and $\Omega_j = \max_g \xi_g \max_g \zeta_g$. The site ratio is

$$\mathcal{R}_j = \frac{(\xi_0\zeta_2 + \xi_2\zeta_0)/2}{\max\{\Psi_j, (1-\delta)\Psi_j + \delta\,\Omega_j\}}, \qquad \delta = \texttt{genotype_replacement_probability} = 0.01.$$

For chromosome z, E_z averages $\prod_{j \in \omega}(1 - \nu + \nu\mathcal{R}_j)$ over `bet_sizes` $\nu \in \{2^{-10}, 2^{-8}, 2^{-6}, 2^{-4}, 2^{-2}, 1\}$ and over windows ω. The windows are the whole contig and half-overlapping windows of `window_widths_bp` = 1, 5 and 20 Mb. E is the mean of E_z over all configured contigs (PS20). A completed-window lower bound may certify a dyad early, and a threshold-straddling interval raises an error (PS21).

Pedigree decision and release

Evidence scale. Let $\mathcal{J}^{\text{tr}}_{cf\mathsf{a}}$ and $\mathcal{J}^{\text{ho}}_{cf\mathsf{a}}$ be the training and held-out log-likelihood sums of configuration a for fold f. Let $\varpi_{c\mathsf{a}} = \log \Pi_{s(\mathsf{a})} - \log n_{s(\mathsf{a})}$, with n_s the eligible configurations in state s. Let $m_{cf\mathsf{a}} \in \{0, -\infty\}$ exclude reciprocal-dominant rows: those naming p as c's parent with $\mathrm{M1}(p \mid c) - \mathrm{M0}(p) \geq \mathrm{M1}(c \mid p) - \mathrm{M0}(c)$, which genetics cannot order. The objective is

$$F(\lambda) = \sum_c \sum_{f=0}^{2} \Big[\log \sum_{\mathsf{a}} e^{\lambda \mathcal{J}^{\text{tr}}_{cf\mathsf{a}} + \varpi_{c\mathsf{a}} + m_{cf\mathsf{a}} + \mathcal{J}^{\text{ho}}_{cf\mathsf{a}}} - \log \sum_{\mathsf{a}} e^{\lambda \mathcal{J}^{\text{tr}}_{cf\mathsf{a}} + \varpi_{c\mathsf{a}} + m_{cf\mathsf{a}}} \Big].$$

Folds are chromosome index modulo 4, with at least two chromosomes per fold. The search evaluates a 33-point grid of $\log\lambda$ on $[\log 0.001, \log 100]$ and refines every eligible interior maximum. All optimisations must succeed, and ties go to the value closest to $\lambda = 1$. The scale is applied only when the gain exceeds 10^{-8}, the optimum lies at least 0.01 from a bound and $\lambda < 1$. Fold 3's test gain is only reported; `evidence_calibration` = "off" sets $\lambda = 1$ (PD03).

Likelihoods and gate. Configuration rows are ordered as M0, then M1 by ascending parent, then canonical panel M2 (PD01–PD02). The M0 mixture uses `m0_contamination_probability` = 0.02 and is applied once (PD04). The gate thresholds are `minimum_edge_exposed_bins` = `minimum_pair_exposed_bins` = 1, `minimum_exposed_fraction` = 0.1 and `minimum_exposed_units` = 3, with `unit_morgans` = 1.0. Units are $\max(1, \text{length}/\texttt{unit_morgans})$, or 1 for a non-finite length. A bootstrap-repeated chromosome counts once. An untestable row keeps the M0 state score and gets an identity score of $-\infty$ (PD05).

Direction and family terms. For a directional log-CDF difference $\Delta_{pc} \neq 0$, with σ the logistic function, chromosome log odds are $\log[(1-\epsilon_{\text{chr}})\sigma(\Delta_{pc}) + \epsilon_{\text{chr}}/2] - \log[(1-\epsilon_{\text{chr}})\sigma(-\Delta_{pc}) + \epsilon_{\text{chr}}/2]$, with `chromosome_error_probability` ϵ_{chr} = 0.05 (PD06). The configuration term is $\log 2 + \log[(1-\epsilon_{\text{pair}})\sigma(O) + \epsilon_{\text{pair}}/2]$, where O is the summed log odds and `pair_failure_probability` ϵ_{pair} = 0.02. A junction-density order probability, a retained heuristic, only orders graph visits, though this can decide which child of a cycle is displaced (`order_minimum_weight` = 2, `order_variance_offset` = 1; PD07). There are `message_passes` = 4 preparatory cavity passes. Rows naming p as c's parent

receive $-\log(1+e^{O^{\mathrm{cav}}_{pc}})$, where O^{cav}_{pc} is the cavity log odds that c is p's parent (PD08). The path factor uses `ancestry_path_budget` = 16, moves a boundary tie class that would overfill the panel to the tail, and is then frozen (PD09).

The final solve runs undamped to `final_tolerance` = 10^{-8} within `final_max_iterations` = 8192, with progress checks every 512 iterations. A stalled solve is retried damped (`retry_iterations` = 4096, `retry_damping` = 0.5) and then resumed. Non-convergence is fatal for the full fit; a resample abstains but stays in the denominator. Release direction log odds are cavity log-odds differences (PD10).

Winners, graph and priors. Two scores tie when their difference is at most $\max(512\,\varepsilon_{\mathrm{mach}}\,\max(1,|\mathrm{score}|),2^{-15})$, under `contrast_ulp_factor` and `noise_floor`. Row index never breaks a tie (PD11). The decision score is the marginal state score plus the identity score minus the state's best identity score, and unresolved-state and unresolved-identity reasons stay distinct (PD12). The acyclic step peels cyclic components; tied weakest children become graph-tie conflicts. Greedy selection then visits children by order probability, confidence and index, and keeps the local row when it is feasible. Otherwise it takes the unique feasible winner from the local state downwards, over `graph_local_search_passes` = 3 passes. This is not global graph optimisation (PD13). The `prior_sensitivity` reselection reuses the family terms and is not repeated in resamples (PD14).

Units and resampling. A chromosome is informative when the child's scaled per-chromosome state evidence, without family terms, has a unique winner. That winner is not required to match the selected state. The chromosome contributes the largest exposed units of the child's eligible edges, or 1 for a child without edges (PD15). Bootstrap uses `bootstrap_replicates` = 1000 draws from a PCG64 stream with `bootstrap_seed` = 20260725. Identical vectors are fitted once and counted by multiplicity. Integer counts make support independent of scheduling (PD16). A precheck needs $N \geq 3$ and 3 informative contigs, and reports attainable support and each tier's minimum contig count (PD17).

Tiers and views. The thresholds (configuration bootstrap, parent bootstrap, LOCO) are `tier_b` = (0.7, 0.7, 0.8) and `tier_a` = (0.95, 0.95, 0.95). The direction cutoff is $\log(0.98/0.02)$, from `orientation_reversal_probability` = 0.02. An exact configuration also needs no graph-tie conflict and, for M0, a graph row that stays M0 under every prior. Unresolved statuses rank prior veto, no stable parent, insufficient support; resolved ones rank graph-tie conflict, graph conflict, unresolved direction, unresolved identity (PD18). Displaced rows of the complete view are never tier-eligible (PD19). A marginal edge needs the same parent in the local and graph rows, both parent supports, no graph conflict and enough units, but the state itself need not pass (PD20). Candidate sets cover `support_set_coverage` = 0.95 of the within-state softmax of decision scores, completing tie classes at a quantum of 2^{-15} (PD21).

Counts, guard and slots. Count-release targets are Tier-B children with an unresolved state or fewer named parents than called. The threshold is $\log(2N/\alpha_{\mathrm{E}})$, with `alpha` α_{E} = 0.01. Release needs every candidate scored, enough units, no graph conflict, at most one survivor, agreement with the supported and complete parents, and no withheld edge, and is labelled as exclusion-based evidence (PD22). The guard uses `minimum_family` = 3, so at

least two tested members, and `member_support` = 0.75. It contracts the edge set to a fixed point in at most $E + 1$ passes, runs count release with the withheld set, and verifies both sets. Guarded views clear the affected calls but keep their candidate lists (PD23). Slots order parents by descending observed-site counts, compared contig by contig, with names breaking exact ties (PD24). Relative-proxy contrasts are diagnostics that no release reads (PD26).

Cost. Sources cost $O(NBS^2)$, transmitted ALT $O(NMS)$, the pair bound $O(R^2BG^3)$ for R retained candidates and $G \le S + 1$, and exposure $O((E/2 + T)BS(S + 1)/2)$ for T admitted pairs. The decision has no intrinsic K term and runs at most $1000 + C$ fits.

Founder allele refit

Evidence uses the founder-construction view, so QC-flagged samples are unobserved. A sample scores at a site only when both of its tracks carry released singleton rows, and sites with no scoring sample are unchanged (FR01). Each site gets `coordinate_sweeps` = 3 sweeps. A sweep has a flip phase, then a two-row phase in which swaps and pair flips are restricted to co-painted rows. A move qualifies when its gain is positive and at least `minimum_gain_nats` = 5 per changed cell. A round applies qualifying moves in decision order, skipping any that share a row, or a row co-painted with one of theirs, with a move already applied. A row changes at most once per phase and sweep, and a sweep without changes ends the site. Qualification and rechecks use the same direct two-row gain (FR02).

Tie classes chain gains within $\max(2^{-16}, (n_i + n_j)2^{-18})$ nats, where n counts scoring carriers (`REFIT_TIE_QUANTUM`, `REFIT_TIE_QUANTUM_PER_CARRIER`). A class wider than `WIDE_QUANTA` = 4 quanta keeps value order; otherwise row or pair index decides. Reversal reorders a class but never changes its membership (FR03). Interacting candidates in one tie class are applied hypothetically in both orders; if either then fails to qualify, all rows involved become unknown in one pass, the round is rescored, and those alleles cannot be filled later (FR04). A fill applies only to alleles unknown in the painted panel. It needs a gain of 5 nats and at least `fill_min_supporters` = 2 carriers whose reads favour the allele by more than 2^{-16}. Fills are decided against the post-sweep state and applied together (FR05). The refit panel goes to family phase and export; pedigree keeps the pre-refit founders (FR06). The cost per site and sweep is $O(N + K^3 \log K)$. In the worst case, two-row gains cost $O(K^4)$, and tie resolution $O(K^4)$ flips and $O(K^5)$ swaps per phase. These analytic worst cases remain under exception 0e (FR04).

Published tables

Painting writes the run-level tables `painting/sample_segments.csv`, `sample_switches.csv`, `founder_support.csv`, `founder_support_windows.csv`, `founder_counts.csv`, `label_continuity.csv`, `painting_model.csv` and `assembly_search.csv`, each with a leading `contig` column. The segment table uses `sample`, `component`, `track`, `start`, `end`, `status`, `label` and `rows`; relatedness and pedigree tables likewise use snake_case names such as `sample`, `parent1`, `parent2`, `parent_state` and `release_evidence`, and retain their run-level paths. These table keys identify inferred output rows and carry no biological parentage truth.

S5. Family phase, recombination maps and released outputs

Family phase

Notation follows the main text. Parameter names are those of the generated parameter reference; values are 4.4.0 defaults.

Emissions, released sites and state alleles (FP01, FP13). The pair emissions $e_{ib}(x, y)$ are the painting's binned emissions, read from its cache or recomputed identically with its `robustness_epsilon` (0.01), `emission_log_floor` (-50 nats) and design effect $1 + (n_{ib} - 1)\varrho_s$. Here n_{ib} counts informative sites of sample i in bin b and ϱ_s is `emission_site_correlation` (0.1). Both solves share these pre-refit emissions. Released sites are refit panel positions inside a painting bin; other sites keep missing calls. A named state's allele $\hat{h}_{xj}$ exists only where every row of its class has a called refit allele and all agree; BACKGROUND has none. Components without painting bins are not solved. They keep their geometry (fallback bins of `EMPTY_COMPONENT_BIN_SITES` = 100 panel sites), but their calls and chain sources stay missing.

Edge rates (FP02). $\theta^{\ell}_b = 1 - \sqrt{\psi_b}$, where ψ_b is the painting's pair stay probability, clipped to $[10^{-15}, (S-1)/S]$. The other clips are $[10^{-12}, 1/2]$ for θ^{m}_b, $[10^{-12}, 1/2]$ for α_b and $[10^{-6}, 1 - 10^{-6}]$ for β_b. `not_copy_rate_per_bp` $\varrho_{\perp} = 10^{-8}$ and `not_copy_mean_bp` $\mu_{\perp} = 5 \times 10^{4}$ give $\omega \approx 5 \times 10^{-4}$. One Haldane implementation (via `expm1`) serves every map distance. `[family_phase] protected_intervals` is a TSV (`chrom, start, end, reason`; 1-based inclusive). Ordinary rows protect their span; a `reversed_segment` row protects its two endpoints. An edge whose closed centre-to-centre interval overlaps a protected interval gets $\theta^{m}_b = 1/2$.

Local gains (FP03, FP25). Because $\perp$ entry draws a uniform label, J is a sum of per-sample terms, each reading only that sample's labels and sources. A move's gain is the accurately rounded sum of after-minus-before terms over exactly the samples it changed (the moved sample and its descendants); no change gives exactly zero. This local gain decides acceptance. The full J, recorded per sweep, is a verification quantity that may round differently near a threshold. A test-only reference path compares the two and never alters production arithmetic.

Block recursion, ties and order (FP04, FP05). The block transition is contracted in three stages (previous source of homologue 0, previous source of homologue 1, previous gauge). This costs $O(B(n_0^2 n_1 G + n_0 n_1^2 G + n_0 n_1 G^2))$, plus $O(B\,|\mathrm{desc}|\,S^2)$ for descendant emissions. Only gauge 0 is allowed at the first bin. In each stage a candidate replaces the incumbent if its score beats the retained best by more than `tie_quantum` (10^{-9} nats), or lies within it and is the slot's current state (for the gauge, the unchanged gauge). The retained argument's score, not the numerical maximum, continues the recursion. At the final bin, the current sources with gauge 0 win ties. Samples are ordered by depth (longest ancestor chain), then index; the graph must be acyclic. These orders fix the Gauss–Seidel updates.

Initialisation and sweeps (FP06, FP07). The first pass solves every sample in index order with both slots free and no child terms, from all-zero labels. The second pass solves samples with a released parent in ancestor order, initially copying parental homologue 0. Neither pass applies a gain test. Sweeps keep a block only if its local gain exceeds

gain_quantum (10^{-6} nats), restoring it otherwise, for at most max_sweeps (30) sweeps. A component is settled when the final sweep of both solves accepted nothing; the recombination summary reports this as family_phase_converged.

Pairing moves (FP08). Candidates are samples with no released parent and at least one child, at every bin edge where both of their homologues change label. They are enumerated once per pass in pedigree then bin order. A move that changes nothing is skipped. A move is kept if its gain exceeds $\max(\log[\tau_s/(1-\tau_s)], \texttt{gain_quantum})$, which is 6.907 nats at $\tau_s = 0.999$. Passes with kept moves trigger more sweeps, for up to max_sweeps passes. At that cap, a final non-keeping pass measures ambiguity in the final state. A pairing is ambiguous when the alternative's gain exceeds $-\log[\tau_s/(1-\tau_s)]$.

Convention solves (FP09). The mirrored solve reverses three orders: visits within a depth, the scan direction for non-incumbent ties, and descendant order during ascent. It keeps the incumbents, gauge preferences, thresholds and first-pass order. Neither solve initialises the other. The default solve's orientation is released even when the run's reporting convention is reversed.

Identity junctions (FP10). A candidate is an edge j and labels $\{a, b\}$ at which at least junction_min_samples (2) distinct samples have a free track, or a track not-a-copy at $j-1$ or j, switching between a and b. A copied homologue's switch belongs to its parent or is a crossover, so it is not counted. A sample carrying $\{a, b\}$ on both sides of j is pinned only under three conditions. A linked homologue must copy parental homologue v at both $j-1$ and j. The parent must not itself carry both labels. On each flank of up to junction_pin_bins (10) bins, stopping where the parent's labels change,

$$\sum_{b'} [e_{pb'}(x, y) - \max_{y'} e_{pb'}(z, y')] \geq \texttt{junction_pin_nats}\,(10),$$

with x the label on homologue v and z the exchanged label. The partner y' is free because two rows exchanged together can fit the same unphased reads. Children never pin: a child's copy excludes the alternative only through a meiosis switch exactly at j, whose odds depend on the interval and on uncertainty the children share. Withheld joins of both solves are combined and cannot be bridged.

Painting ties and forced breaks (FP11, FP12). At each boundary between a sample's released painting chunks, direct and swapped continuations are compared lexicographically by (label changes, allele-visible changes). The boundary is a tie unless direct continuation is strictly better. It is mapped to the next site's bin and kept only at internal bins of samples with no released parent. Forced breaks combine tie bins, withheld joins and ambiguous-pairing edges (applied to the parent and all its children) from both solves. Only tie-only cuts can be bridged.

Panel-allele uncertainty (FP14). The prior is the logit of the chromosome-wide share of allele 1 among called refit cells, clipped to [0.01, 0.99] (PRIOR_BOUNDS). Carriers of x at site j are samples whose optimal label pair contains x; samples flagged by sample quality control are excluded, though they are still phased. Each carrier adds $\ell(g_1) - \ell(g_0)$, where $\ell(g) = \log[(1-\varepsilon)P(r_{ij}, a_{ij} \mid g) + \varepsilon/3]$, ε is the painting's robustness_epsilon, and g_u is the carrier's dosage if x carries allele u given its partner's selected allele; an (x, x) carrier compares dosage 2 with 0. With $\gamma_{xj} = \text{logistic}(\text{prior} + \Lambda_{xj} - \Lambda_{xj}^{(i)})$, where Λ_{xj} sums the carrier terms, d_{xj} is the disagreement of γ_{xj} with the selected allele, capped at 1/2 (FLIP_CEILING)

and zeroed below 10^{-15}. The sample's own direct term $\Lambda^{(i)}_{xj}$ is removed for the labels of its optimal pair. Documented approximations remain. This is not a cross-fitted refit, and the sample can still influence carrier and panel selection. The two copies flip independently even for (x, x), and correlated carrier identity errors remain a failure mode.

Posterior, site model and calls (FP15–FP17). Forward–backward uses the Viterbi factors, starts the gauge at 0 and enforces any anchor in both directions. States with log posterior ≤ -700 are dropped without renormalisation. Descendant sums follow a fixed (depth, index) order in both solves. Chain probabilities stay in float64 until the release test. Settings are refused unless $\tau_s < (1 - 5\delta/6)/(1 - 2\delta/3)$, the orientation certainty of a fully determined heterozygote (0.99983 at $\delta = 10^{-3}$). They also require $\tau_p \leq \tau_s$, all thresholds in $(1/2, 1)$ and $\delta \in (0, 1/2)$. `site_deviation` (δ) and `set_posterior` (τ_s) are working values chosen on validation simulations, not estimated error rates. Genotypes use the argmax dosage, with the lower dosage winning exact ties, at the inclusive `genotype_posterior` ($\tau_g = 0.9$). Heterozygotes are phased at the inclusive `phase_posterior` ($\tau_p = 0.99$). The order is (1,0) only if $P_{10} > P_{01}$, otherwise (0,1). Calls carry both alleles or neither.

Gauge-sample sets (FP18). These rules apply to samples with neither a released parent nor children. The anchor is the first bin of a partition whose optimal labels differ and whose label pair is heterozygous with posterior $\geq \tau_p$. Certainty is computed over whole bins up to the next forced break. The set stops at the first later bin with differing optimal labels, label heterozygosity $\geq \tau_g$ and certainty $< \tau_s$. Undefined certainty (no confidently heterozygous site) does not stop it. Pre-anchor bins join the anchored partition; each stop opens a new partition.

Children-anchored gate (FP19). For a sample without released parents but with children, a tie-only cut is bridged when the certainties at the nearest informative bins before and at-or-after the cut both reach τ_s. A cut lacking such a bin on one side is kept. A break kept by either solve is kept.

Agreement (FP20). Breaks are combined, and a set also starts wherever the relative orientation of the two solves changes between consecutive phased heterozygotes. The first site is never a break.

Chain release (FP21). Resets fall at the component start, wherever $\theta^m_b \geq 1/2 - 10^{-12}$, and at the bins of either individual's agreed set starts. A start inside a bin (after its first released site) blocks that bin and resets the next. $\kappa_{eb}(v)$ excludes not-a-copy mass. For a parent without a released parent, alignment uses the bins of each reset segment where the default label pair is heterozygous and the unordered pairs match. If all those bins are swapped, the mirror is flipped; if only some are, the segment is blocked. Bins whose aligned parental labels still differ are blocked. Supports are stored as float32, zero where unreleased.

Storage and product (FP22–FP24). Resident or windowed storage, fixed 4,096-site chunks and worker count do not change calls, phase sets or chains. Each product binds the exact parent-edge digest it was conditioned on. Painted-span coverage is diagnostic, and recombination does not read it.

Cost in K. A sweep costs $O(B(NS^3 + S^2 \sum_i A_i))$, with A_i one plus i's descendant count. A pairing move costs $O(n_c BS^3)$ for n_c children. Release costs $O(NB(S+2)^3 G)$ for posteriors, $O(LS^2)$ per sample for the site model, and $O(EB)$ for chains. Emissions take $8NS^2 B$ bytes. Every term is at most cubic in K ($S \le K + 1$).

Recombination maps

Breakpoints (recombination RC01, RC02). `[recombination] order_breakpoints` uses the protected-interval TSV format; the two settings are independent. Ordinary intervals (in file order), then reversed-segment endpoints as point intervals, are made half-open as $[\text{start}, \text{end} + 1)$. Each is widened cumulatively to every overlapping painting bin and removed from the map axis, with a new segment after it. Inversion interiors are not masked automatically, and calls inside reversed segments are flagged. A chain bin spanning two segments or a masked zone is invalid. Segment changes reset chains.

Following (recombination RC03). A marker is a released bin in a valid segment; stretches restart at any chain or segment reset. A called interval is $(c_{m-1}, c_m]$, between bin centres. Its reported interval is the inclusive bp range of both bins, so a change inside either bin is not mislocated. The joint lower bound $\max(0, \bar{\kappa}_l + \bar{\kappa}_r - 1)$ holds under any dependence. For each meiosis the stage counts released, followed and unresolved bins.

Conditional map (recombination RC04, RC05). Map bins start every `bin_bp` (1,000,000 bp) from the first site, with extra edges at the two sites flanking each segment change. λ is the difference of cumulative Morgans interpolated at bin centres. Map knots between the end sites are retained, so hotspots between SNPs keep their mass. The fallback rate `[genetic_map] rate_cm_per_mb` (default 5 cM/Mb) applies without a map and beyond its knots. An odd interval of zero genetic length contributes one crossover: the data override the map's zero prior. Edge sums follow a fixed order. The columns are:

- `rate_c_m_per_mb` is $10^8 X_t / W_t$, missing where $W_t = 0$.
- `covered_bp` is the union of all meioses' informative intervals within the bin.
- `c_m_on_covered_bp` is rate × covered bp / 10^6; the cumulative sum counts missing values as zero.
- `effective_meioses` is W_t divided by the map-bin width.
- The called rate divides called crossovers (spread by genetic mass, else uniformly in bp) by callable exposure.

`rate_is_measured` does not alter the calculation.

Junction diagnostic (recombination RC06). Overlapping or touching reported intervals form loci, and a locus is admitted when at least `minimum_parents` (3) distinct parents have calls in it. Each edge is scored by the parity of its calls in the locus. It is decidable if it has a call there or a callable span covering the calls' extent. A parent with η_1 odd of η decidable meioses contributes $\log\left[\frac{1}{2} q^{\eta_1} (1-q)^{\eta - \eta_1} + \frac{1}{2} q^{\eta - \eta_1} (1-q)^{\eta_1}\right]$. Here q is the probability of a real transmission change, and the mixture is over a frame flip (prior 1/2) shared by the parent's children. Three statistics follow.

- **Mixing.** The maximum over 500 values $q \in [0.001, 0.5]$, minus the maximum over $q \le q^{\max}$. The ceiling is $q^{\max} = q_{\text{pt}} + r_{\text{bg}} - 2 q_{\text{pt}} r_{\text{bg}}$, with `hotspot_switch_probability` $q_{\text{pt}} = 0.05$ and r_{bg} the Haldane fraction across the locus.

- **Pairing.** For each locus pair, a discordance mixture over 50 values $\phi \in [0.005, 0.25]$, among parents decidable at both loci, minus the larger independent switching maximum. The best partner is kept, minus $\log$ (locus pairs tested).
- **Flag.** A locus is flagged when the larger statistic reaches $\log 100$ (`flag_log_likelihood_ratio`).

The documented rationale is that no locus gives real crossovers in more than 5% of meioses beyond background, with flags at 100:1. Ties take the first maximum, or the last under reversed conventions. The ratios are conditional on the calls, not calibrated odds, and never mask the map.

Stage and cost (recombination RC07). The stage refuses chains conditioned on another parent-edge digest. Its cost has no intrinsic K dependence: $O(EB)$ following over the B chain bins, $O(EB_m + \text{calls})$ aggregation over B_m map bins, and a detector cost of $O(\text{loci}^2\, E)$ plus $O(\text{loci}^2 \times \text{parents} \times \text{grid})$.

Released outputs

Site rows (OU01). By default (`[export] full_length = true`) export writes every input biallelic SNP reaching the site filter, in position order. Multiallelic, non-SNP and duplicate-position records dropped by the loader are absent. Kept sites have FILTER `PASS`; removed sites list their rules in r1–r5 order; a kept-only export writes `.`.

Founder VCF (OU02–OU04, OU06). Column `cC_Hk` holds row k of release segment C as a haploid GT, and is `.` outside the segment's panel span. Upstream (OU03), a called allele is withheld unless an eligible painted carrier supplies observed read support, using refit calls. At sites removed only by the informativeness rule r5, unwithheld rows receive the strict majority of summed REF and ALT allele depths (GT only where AD is unusable). Ties stay `.`, and the fill carries no founder-specific evidence. Sites removed by r1–r4 stay `.`. Founder-identity flags mark duplicate-partner changes between blocks, and are diagnostic only.

Sample VCF (OU05, OU09). Columns are in input order. A set starts at a component's first kept site and at the site after each released break; its PS is that bp position, which need not be heterozygous. Inside a set, homozygous and missing cells also use `|` and PS. A frame-undetermined heterozygote is written `/` with ascending alleles (missing first) and PS `.`, without ending its set. Sites outside components are unphased, removed sites are `./.:.`, and there is no r5 fill.

Recombination tables (OU18). Run-level `recombination/crossovers.csv`, `meioses.csv`, `map.csv`, `coverage.csv`, `junctions.csv` and `summary.csv` use a leading `contig` column and snake_case fields; call intervals are $(\text{left}, \text{right}]$. The single coverage table contains spans between consecutive released markers within a followed stretch, used by map aggregation, called exposure, junctions and evaluation. `recombination/maps.pdf` has one page per chromosome in configured contig order, presenting the same conditional map as the tables.

Identity and optional tracks. `export/founder_identity.csv` is the run-level identity table, with `contig`, `component` and `founder` identifiers. Optional per-chromosome `export/<contig>.tracks.csv.gz` files use `contig`, `component`, `pos` and `hk` columns; shorter

segments leave unused founder columns empty, while an existing unknown allele remains . . The per-chromosome VCF paths and their sample/founder column identifiers are retained.

Cost. The sample VCF costs $O(NL)$ and the founder VCF $O(L \times \text{columns})$ over L written sites, at most quadratic in K.

S6. Simulation, evaluation and computation

Simulation settings and constants

The following table lists every simulation setting with its default. Settings in `[simulation]` and `[simulation.reads]` describe the generating model only; inference keeps its own read model and map. The order of `[input] contigs` is part of the random streams, and every contig's template must hold the same even number H of haplotypes.

Setting	Default	Allowed values	Meaning
`seed`	400	integer or `"random"`	Seed; `"random"` takes eight OS-random bytes once at initialisation and records the realised value
`templates`	none (required)	directory of `<contig>.npz`	`positions` int64[L], `allele_probabilities` float64[H, L, 2]
`cohorts`	[20, 100, 200]	each ≥ 2	Cohort sizes; cohort 1 descends from the founders
`depth`	5.0	finite, > 0	Mean depth λ
`generating_rate_cm_per_mb`	5.0	—	Fallback rate γ (5 cM/Mb $= 5 \times 10^{-8}$ Morgans per bp)
`generating_map`	none	cumulative cM map	Generating map, independent of inference
`backcross_fraction`	0.0	$[0, 1]$	Per-child backcross probability in eligible cohorts
`backcross_start_cohort`	3	≥ 2	First cohort that may backcross
`observe_cohorts`	[] (all)	unique, generated cohort numbers	Cohorts eligible for sequencing
`observed_fraction`	1.0	$(0, 1]$	Sequenced fraction θ of eligible individuals
`pedigree`	none	TSV: sample, parent1, parent2, sex, generation, observed	Explicit pedigree; replaces `cohorts` and random matings; `backcross_fraction`, `observe_cohorts` and `observed_fraction` must keep their defaults
`reads.error_rate`	0.02	$(0, 0.5)$	ε

Setting	Default	Allowed values	Meaning
reads.depth_cv	0.0	finite, ≥ 0	v
reads.heterozygote_alt_probability	0.5	$(0, 1)$	b
reads.dropout_fraction	0.0	$[0, 1)$	ϕ
reads.read_length_bp	0	integer ≥ 0	ℓ; $\ell > 0$ selects fragments
reads.fragment_mean_bp	375.0	> 0 when $\ell > 0$	μ_{frag}
reads.fragment_sd_bp	150.0	≥ 0 when $\ell > 0$	σ_{frag}

The next table lists the internal constants, which are not user settings.

Constant	Value	Role
MUTATION_RATE_PER_BP	10^{-10}	μ, per bp per meiosis
MINIMUM_FRAGMENT_BP	50	Lower clip of insert length
FRAGMENT_SD_LIMIT	6	Upper clip $z_{\max} = \lceil \mu_{\text{frag}} + 6\sigma_{\text{frag}} \rceil + 1$ bp
FRAGMENT_QUANTILES	100,000	Quadrature points for the expected bases covered per fragment
WINDOW_BP, WINDOW_SHIFT_BP	100,000; 50,000	Template windows and stride
TEMPLATE_HAPLOTYPES	6	Linker path budget
UNDECLARED_LENGTH_BP	300,000,000	Contig length when the VCF header gives none
SIMILARITY_CUTOFF	50 (%)	Junction mismatch cutoff
NODE_PENALTY, EDGE_PENALTY	10; 10	Penalties applied after each extracted path
GAP_SCORE, EDGE_DIVISOR	1; 3	Score of non-overlapping windows; edge weight = score / 3
BISECTION_STEPS	60	Inversion of merged information
PATH_TOLERANCE, CANCELLATION_TOLERANCE	10^{-9}	Path reconstruction; exact information cancellation

The validation configurations are summarised below. The example generator illustrates software wiring and is not an accuracy experiment.

Validation datasets and configurations

The matrix introduced for 4.2 and rerun for 4.4 contains 39 runnable entries; rows retained only as historical, non-runnable matrix entries are excluded. Fifteen distinct simulated datasets have truth: the thirteen design cases and the two references named zero_std_n80_ref and zero_stress_n80_ref. Their other variants repeat inference on the same generated data. n80 and stress abbreviate those two reference names in the manuscript. Each table row is one configured seed, not a replicated factorial cell. std and seed repeat the same cohort design under different seeds.

Supplementary Table S3. Simulation matrix. N is observed samples; H is true founder haplotypes, distinct from the inferred component count K; depth is mean read depth; v is between-sample depth CV, ε homozygous read-error probability, b heterozygous ALT probability, ϕ dropout fraction, and BC is backcross probability from cohort 3. All inference fallback rates are 5 cM/Mb and marked unmeasured. “Knots” denotes the explicit map described below.

A. Cross design and observation.

Scenario	Seed	N	Chr.	H	Generated cohorts	Observed cohorts	θ
std	9101	40	8	6	10,15,15	all	1.0
seed	9217	40	8	6	10,15,15	all	1.0
n80	9101	80	8	6	20,30,30	all	1.0
backcross	6202	80	3	6	20,30,30	all	1.0
bc8	6202	80	8	6	20,30,30	all	1.0
longbc	32011	80	3	6	20,30,30	all	1.0
f3	9603	112	3	6	16,4,96	1,3	1.0
longf3	32012	112	3	6	16,4,96	1,3	1.0
f3x8	9608	112	8	6	16,4,96	1,3	1.0
sibrich	32001	112	8	6	16,4,96	1,3	1.0
tl	2171	115	8	6	explicit: 3, 34, 96	table flags	1.0
famhs	32002	60	8	6	explicit: 3, 12, 38, 9	table flags	1.0
hsrand	32003	66	8	6	6,60	all	1.0
stress	9102	64	8	6	20,30,30	all	0.8
k30	9101	80	5	30	60,20	all	1.0

B. Read model, backcrossing and maps.

Scenario	Depth	v	ε	b	ϕ	Read bp	BC fraction	Generating cM/Mb	Inference map
std	5.0	0.0	0.02	0.5	0.0	0	0.0	5.0	fallback
seed	5.0	0.0	0.02	0.5	0.0	0	0.0	5.0	fallback
n80	5.0	0.0	0.02	0.5	0.0	0	0.0	5.0	fallback
backcross	5.0	0.0	0.02	0.5	0.0	0	0.5	5.0	fallback
bc8	5.0	0.0	0.02	0.5	0.0	0	0.5	5.0	fallback
longbc	5.0	0.0	0.02	0.5	0.0	0	0.5	5.0	knots
f3	10.5	0.25	6.7e-05	0.5	0.0	125	0.0	2.0	fallback
longf3	10.5	0.25	6.7e-05	0.5	0.0	125	0.0	5.0	knots
f3x8	10.5	0.25	6.7e-05	0.5	0.0	125	0.0	2.0	fallback
sibrich	10.5	0.25	6.7e-05	0.5	0.0	125	0.0	2.0	fallback

Scenario	Depth	v	ε	b	ϕ	Read bp	BC fraction	Generating cM/Mb	Inference map
tl	5.0	0.0	0.02	0.5	0.0	0	0.0	5.0	fallback
famhs	5.0	0.25	0.02	0.5	0.0	0	0.0	5.0	fallback
hsrand	5.0	0.25	0.02	0.5	0.0	0	0.0	5.0	fallback
stress	5.0	0.6	0.04	0.6	0.1	0	0.5	5.0	fallback
k30	5.0	0.0	0.02	0.5	0.0	0	0.0	5.0	fallback

The eight-chromosome set for `std`, `seed`, `n80`, `stress`, `bc8`, `tl`, `famhs` and `hsrand` is chr1, chr2, chr8, chr10, chr13, chr18, chr19 and chr20. The eight-chromosome omitted-parent set (`sibrich`, `f3x8`) replaces chr18 by chr9. All three-chromosome designs use chr2, chr9 and chr19. The $H = 30$ design uses five short template contigs m1–m5, each with 8,000 sites, with spans of 634,611, 468,400, 586,577, 627,850 and 511,719 bp. Its template-construction seed is 1, separate from the pedigree/read seed 9101.

The empirical-template manifest records six haplotypes per chromosome, source discovery products and content hashes. It does not identify the original biological sampling or input VCF beyond those products. The $H = 30$ manifest records crops sourced from chr19, chr8, chr13, chr20 and chr10, respectively; these are a constructed count stress test rather than thirty documented biological founder sequences. Template probabilities enter the simulator as described above.

For cohort designs, the listed cohort sizes exclude the unobserved template founders. The omitted-parent designs generate cohorts (16, 4, 96), sequence only cohorts 1 and 3, and therefore have no observable parent edges. The explicit `tl` table has 133 generated individuals (3, 34, 96 by its descriptive generation groups), with 115 observed; `famhs` has 62 (3, 12, 38, 9), with 60 observed. Those tables' observed flags replace random thinning. Sex and generation columns are simulation truth descriptions only. Every other row uses the cohort sampler and its configured observation fraction.

Fragment simulations use 125 bp reads with insert mean 375 bp and SD 150 bp. Independent-site simulations set read length to zero. All rows not using fragments retain the same dormant insert settings. The stress reference combines $v = 0.6$, $b = 0.6$, $\phi = 0.1$, $\varepsilon = 0.04$, backcross fraction 0.5 and observation fraction 0.8; it does not isolate their effects.

All generating maps are uniform fallbacks at the rates in Table S3, with no generating map file. `longbc` and `longf3` use an explicit inference map with first knots at position 1 and 0 cM, and last knots (bp, cM) of (39,628,896, 198.144480) on chr2, (37,982,278, 189.911390) on chr9, and (31,954,172, 159.770860) on chr19. Thus its interior slope is 5 cM/Mb and its knots provide map length to the release rules, unlike an unmeasured fallback. These experiments change map support, not just crossover rate; we use the recorded knots as a synthetic map-support input and make no claim of empirical provenance.

Supplementary Table S4. Phase by supported parent edges. Rows group samples by the actual post-guard edge product used by phase, including marginal edges. They are not groups defined by true parents or cohorts. A zero denominator is undefined; the

corresponding N50 is shown as a dash. Spans and ratios pool chromosomes within one run only.

Scenario	Supported parent edges	*N*	Genotype errors / calls	Phased / correct true heterozygotes	Switches	Long switches	Released set N50 (Mb)
std	0	10	8007/23167772	4912889/5179823	5	1	36.315
std	1	0	0/0	0/0	0	0	—
std	2	30	8776/70806131	13213171/13464156	14	0	36.316
seed	0	10	4342/23382342	4652196/4856587	8	0	32.399
seed	1	0	0/0	0/0	0	0	—
seed	2	30	6263/70831948	12760682/12917017	14	0	36.295
n80	0	20	2787/47220303	10318693/10529198	4	0	36.310
n80	1	0	0/0	0/0	0	0	—
n80	2	60	12983/141719342	27030210/27351284	10	2	36.316
backcross	0	80	5792/77938894	14208601/14358269	1	1	4.716
backcross	1	0	0/0	0/0	0	0	—
backcross	2	0	0/0	0/0	0	0	—
bc8	0	20	2268/47244076	10055605/10188412	0	0	35.621
bc8	1	0	0/0	0/0	0	0	—
bc8	2	60	13160/141719151	25499544/25882411	3	0	36.298
longbc	0	20	815/19480748	3930447/3960588	0	0	31.953
longbc	1	2	146/1948346	267288/277657	0	0	31.953
longbc	2	58	4312/56501244	9469672/9673075	1	1	37.599
f3	0	112	826/109152120	18652911/18797686	21	0	17.508
f3	1	0	0/0	0/0	0	0	—
f3	2	0	0/0	0/0	0	0	—
longf3	0	112	1781/109084847	20064580/20251987	8	2	6.103
longf3	1	0	0/0	0/0	0	0	—
longf3	2	0	0/0	0/0	0	0	—
f3x8	0	112	3078/270232710	50025591/50431364	127	2	11.523
f3x8	1	0	0/0	0/0	0	0	—
f3x8	2	0	0/0	0/0	0	0	—
sibrich	0	112	2551/270316348	50976343/51517801	23	2	15.518
sibrich	1	0	0/0	0/0	0	0	—
sibrich	2	0	0/0	0/0	0	0	—
tl	0	61	11112/144119947	27580738/27895961	3	1	7.299
tl	1	28	5118/66146713	12242056/12588579	0	0	31.949
tl	2	26	3474/61422994	11981561/12168306	2	0	35.620
famhs	0	11	2987/25960612	5635031/5765681	226	6	10.328
famhs	1	0	0/0	0/0	0	0	—
famhs	2	49	7435/115712051	22389628/22574200	8	0	36.316

Scenario	Supported parent edges	*N*	Genotype errors / calls	Phased / correct true heterozygotes	Switches	Long switches	Released set N50 (Mb)
hsrand	0	6	303/14170639	3203486/3228751	0	0	36.319
hsrand	1	0	0/0	0/0	0	0	—
hsrand	2	60	7454/141724778	26911473/27259539	0	0	36.316
stress	0	21	24939/49396786	9073845/10282058	3	3	28.334
stress	1	18	40206/42196883	6540485/7344465	9	9	32.678
stress	2	25	56854/58782098	9961245/10483334	21	11	35.492
k30	0	71	1285/2736202	335119/345872	31	0	0.586
k30	1	7	71/278664	30879/32574	0	0	0.586
k30	2	2	20/79206	9875/10059	0	0	0.587

Supplementary Table S5. Supported marginal-edge product against simulation truth. Only named, supported edges in the partial Tier-B table count. True parents outside the sequenced sample set are excluded. This is the edge product used by phase and recombination, and can contain more edges than the exact-configuration call product scored in main-text Table 3.

Scenario	Supported edges	Correct supported edges	Extra supported edges	Missing supported edges
std	60	60	0	0
seed	60	60	0	0
n80	120	120	0	0
backcross	0	0	0	120
bc8	120	120	0	0
longbc	118	118	0	2
f3	0	0	0	0
longf3	0	0	0	0
f3x8	0	0	0	0
sibrich	0	0	0	0
tl	80	80	0	0
famhs	98	98	0	0
hsrand	120	120	0	0
stress	68	68	0	0
k30	11	11	0	29

TroMau chromosome summaries

Supplementary Table S6. TroMau QC and founder release. SNP rows are those written by the full-length export, including rows removed by site filtering; kept sites are the inference set. Blocks are 200-site discovery blocks, including the final partial block. Sample QC flags affect founder construction and do not remove those samples from painting or pedigree inference. Release segments are counted from the export manifest; they need not equal the number of source assembly components. Founder labels reset between segments.

Chromosome	SNP rows	Kept sites	Blocks	Flagged samples	Release segments	Founder rows per segment
chr1	233533	93460	468	1	1	8
chr2	254453	89047	446	1	1	8
chr3	749986	128257	642	1	1	8
chr4	248773	81564	408	1	1	8
chr5	232844	87364	437	1	1	9
chr6	269001	97524	488	1	1	8
chr7	422133	166616	834	1	1	8
chr8	145568	58101	291	1	1	9
chr9	286666	80058	401	1	1	8
chr10	210484	83637	419	1	1	9
chr11	233185	90287	452	1	1	7
chr12	253473	86871	435	1	1	8
chr13	195764	76025	381	1	1	9
chr14	246481	92774	464	1	1	9
chr15	229857	90025	451	1	1	7
chr16	218895	78525	393	1	1	8
chr17	236718	94486	473	1	1	8
chr18	255678	83141	416	1	1	8
chr19	210916	71342	357	1	1	8
chr20	204509	74372	372	1	1	8
chr22	261928	89376	447	1	1	8
chr23	318039	102315	512	1	3	8,8,8

The 320-sample depth series

The separate series uses `configs/simulation.toml` at release 4.4.0 (commit `867f163`), seed 400 and the empirical six-haplotype templates. All 320 animals in cohorts [20, 100, 200] are observed; their template founders are unobserved. There are 22 contigs: chr1–chr20, chr22 and chr23, in that order. Read depth is the only changed generating parameter (1, 1.5, 2 or 5). Reads use the default independent-site Poisson model, error probability 0.02, heterozygous ALT probability 0.5, no depth variation and no dropout. Generating and inference fallback rates are both 5 cM/Mb, with no map files and the inference fallback marked unmeasured. Backcrossing is off. Depths 1 and 1.5 additionally set the execution-only `memory_limit` to `250G`, which the runtime interprets as 250 GiB. The runs use one seed and one replicate per depth; no cross-depth hardware or lane-count matching was performed.

Supplementary Table S7. The full depth series. Panel A gives rates; panel B gives exact final founder-refit and genotype counts; panel C gives phase and pedigree; panel D gives computation. Main-text Table 6 reports the corresponding rates. All founder rows are matched, so using matched called alleles (the manuscript definition) or all called alleles (the comparison script's definition) gives the same one-to-one error rate here. Founder missing rates divide missing by called plus missing alleles; the sum is 53,741,112 at every depth. Genotype rates divide errors by called genotypes. Set-aligned

site errors count minority-orientation sites within each released set and divide by phased true heterozygotes; they are not switch events. Phased share divides those heterozygotes by correctly called true heterozygotes. N50 is the global released-set span N50 from evaluation totals, recomputed from pooled spans by the evaluator, not averaged over chromosome N50s.

A. Rates and release.

Depth	Founder wrong / missing per 10^6	Genotype errors per 10^6	Set errors per 10^6	Phased share
1	5.55 / 19.05	115	68	0.9976
1.5	1.51 / 5.34	107	66	0.9976
2	0.54 / 2.20	98	50	0.9975
5	0.20 / 0.02	63	60	0.9913

B. Founder and genotype counts.

Depth	Called founder alleles	Wrong	Missing	Called genotypes	Genotype errors
1	53,740,088	298	1024	2,864,927,399	330,862
1.5	53,740,825	81	287	2,864,738,842	305,481
2	53,740,994	29	118	2,864,497,726	280,489
5	53,741,111	11	1	2,864,012,489	179,900

C. Phase and pedigree counts.

Depth	Phased / correct true het.	Set site errors	Switches / long	Released sets	Released-set N50 (bp)
1	487,466,845 / 488,652,509	32,979	31 / 13	7,101	39,779,055
1.5	487,541,151 / 488,701,752	32,028	27 / 7	7,118	39,779,037
2	487,471,374 / 488,707,602	24,468	20 / 6	7,114	39,779,055
5	484,309,497 / 488,570,031	29,210	19 / 7	7,143	39,759,093

Every row has 600/600 true edges correct, no extra or missing edge, 320/320 states correct and 320/320 exact configurations. The expected states are 20 zero-observed-parent and 300 two-observed-parent states; no sample has exactly one observed parent.

D. Run-command computation.

Depth	Run (h:mm:ss)	Peak live (GiB)	Limit (GiB)	GNU max. RSS (GiB)	Processor / cores
1	5:56:02	225.2	250.0	4.91	Sapphire Rapids / 112
1.5	5:46:27	225.9	250.0	5.81	Sapphire Rapids / 112
2	7:34:09	176.0	249.9	5.42	Ice Lake / 76
5	4:44:29	254.8	500.4	5.34	Sapphire Rapids / 112

Peak live and limit columns convert `memory.peak_live_use_bytes` and `memory.limit_bytes` from the completed `run` record using 2^{30} bytes/GiB. The peak is sampled job memory pressure, including concurrent workers and possible node pressure, excluding reclaimable inactive file cache. It is neither a sum of worker RSS nor a minimum required capacity. The limit is the recorded governor bound, which need not equal a nominal node label. The two configured 250 GiB limits were applied on larger nodes; depth 2 used the nominal 250 GB node. GNU time's maximum process RSS converts KiB using 2^{20} KiB/GiB and covers `run` only. Its value is distinct from the maximum over the separate init, run and evaluate commands in the campaign comparison helper. Wall time is GNU time's elapsed `run` command, including command overhead, with init and evaluation excluded; the runtime's narrower internal wall timer is not substituted. All four runs completed in one attempt, with zero memory waits and zero forced grants. Run admissions were 422,645, 402,551, 380,994 and 301,699 in increasing depth order. No timing or memory uncertainty can be estimated from these single observations on mixed hardware.

Generating-model details

Templates (OU20, OU21). Windows start every 50,000 bp from 0 to the contig length and cover $[\text{start}, \text{start} + 100,000)$. For rows k and k' of consecutive windows, the mismatch percentage Δ is 100 times their expected allele mismatches, $\sum_j P_k(0)P_{k'}(1) + P_k(1)P_{k'}(0)$, divided by the number of shared discovery sites; the similarity $\sigma = 0$ if $\Delta > 50$, otherwise $2(50 - \Delta)$, and the junction score is $100(\sigma/100)^2$. Paths are found by reverse dynamic programming, the first candidate winning exact ties. After each path, every row's node score in a used window falls by 10 times its similarity to the used row, with similarity $1 - \min(1, 2\delta/n)$ for δ expected mismatches over the window's n discovery sites, normalised by the square roots of the diagonal, and each internal used edge falls by 10. Overlaps are merged with signed information $I(p) = \text{sign}(p - \frac{1}{2})(1 - H_2(p))$, where H_2 is the binary entropy, combined as $(I_1 + I_2)/(1 + I_1 I_2)$ and inverted by bisection. A contig with fewer than two informative windows gets no template. The linker is compiled without fast-math reassociation (ledger L14), which matters only when templates are rebuilt.

Founders (OU23). Each haplotype makes one vector draw for its exact ties, in site order, and only when ties exist.

Pedigrees (OU24, OU25). In the cohort design, the master stream's first draw seeds the structural generator. Each cohort's structure is built before its seed block is drawn. In a backcross-eligible cohort every child consumes one structural uniform draw: below `backcross_fraction`, the first parent is drawn uniformly from the previous cohort and the

second uniformly from that parent's two parents; otherwise two distinct previous-cohort members form an ordered pair (first parent gives the maternal haplotype). In an explicit table, founders have depth 0 and others one more than their deeper parent; individuals are simulated by depth in table order, without a structural draw, and the founder count must equal $H/2$. Generated identifiers (`Founder_i`, `F<cohort>_<i>`) are truth labels only.

Meiosis (OU26). Without a map, gaps of $\lceil \mathrm{Exp}(\text{mean } 1/\gamma) \rceil$ bp run from the first marker until past the last. With a map, the Poisson mean is the distance from first to last marker, including fallback-rate tails beyond the map's knots. A zero distance or rate consumes no crossover draws, and a marker exactly at a crossover takes the new homologue. Mutation gaps are $\lfloor \mathrm{Exp}(\text{mean } 1/\mu) \rfloor$ bp; each event flips the first marker at or after it, and duplicate indices flip once. The child stream draws the maternal gamete, then the paternal.

Reads (OU27, OU28). The independent-site sampler draws, in order: depth factors (only if $v > 0$), the full $N \times L$ Poisson array, dropout starts (only if $w > 0$), then binomial ALT counts per dosage class 0, 1, 2, skipping empty classes. The fragment sampler draws factors once. Then, per sample, it draws a Poisson fragment count over the span from $z_{\max}$ bp before the first marker to the last marker, uniform starts, lengths and homologues, followed by per-read imbalance and error draws. Dropout starts are drawn after all samples. The start rate is $\lambda u_i/(1 - w/L)/\bar{c}$, where $\bar{c}$ is the mean of $\min(z, 2\,\ell)$ over insert lengths z at the quadrature points. Both samplers consume whole-contig arrays, so splitting by sample would change the draws.

Streams (OU30).

Stream	Construction	Draws
founders	PCG64, `SeedSequence([seed, 2000000])` child c	Tie resolution on contig c
master	PCG64, seed	Structure seed (cohort design), then per cohort or depth a contig-major $C \times n$ child-seed block
structure	Mersenne Twister seeded by `master.integers(0, 2^31)`	Parent pairs and backcross choices
child	PCG64, child seed	Maternal then paternal gamete
observation	PCG64, `SeedSequence([seed, 3000000])`	Sequenced subset
reads	PCG64, seed + 1,000,000	One seed per contig, for every contig
contig reads	PCG64, contig read seed	Whole-contig reads

Here C is the number of contigs and n the cohort size.

Truth (OU29). Per contig: positions; founder haplotypes int8[H, L] (consecutive row pairs form the diploid founders); sequenced samples' alleles int8[N, L, 2] (maternal, paternal); ancestry chunks with integer $[\text{start}, \text{end})$ bounds and maternal and paternal founder haplotype (-1 for none); and one row per gamete of a sequenced child (child, parent, slot, cohort index, float crossover positions). The run manifest holds the state,

requested and realised seeds, contigs, source identity, sequenced sample identifiers, and the sequenced and generated pedigree tables. `simulation/truth_pedigree.csv` summarises the sequenced samples' true parents.

Metric definitions and edge cases

Selection (OU31). The default `available` mode scores the latest founder product of each contig plus any completed pedigree, family phase and recombination products, listing missing downstream products as skipped. `all` scores every founder product present; explicit stages fail if a product is missing. A complete simulation manifest is required. The pedigree comparison stays genome-wide when contigs are restricted.

Founder panels (OU32, OU33). A stage's components must not overlap and must lie on truth markers, and every truth marker counts. If $K_c > H$, $K_c - H$ rows stay unmatched and their calls are reported as unmatched called alleles; if $K_c < H$, each unmatched founder costs L_c missing alleles. When several assignments are equally optimal, assignment-derived counts and strata can differ between them; a pure row permutation changes no count. Per-million rates are null with a zero denominator. In the segment path, an equal switch or stay cost keeps the current label, and the first terminal minimum wins. An infinite switch cost reproduces the nearest-founder count. Component-wise counts can fall when a component is cut, because labels re-match on each piece; the split-charged cost rises by at most κ per row.

Ancestry and strata (OU34). Founder f is represented at site j if $c_{fj} \geq 1$, ignoring unknown ancestry. Chunks are half-open, except that the chunk ending at the last marker includes it. Per stratum, missing alleles are truth alleles minus correct and wrong matched calls, so they include uncalled alleles, unmatched founders and uncovered markers.

Identity continuity (OU35). Windows are consecutive blocks of 100 truth markers, the last possibly partial, each centred on its middle marker. A row is labelled in a window when it has $n \geq 16$ calls there, the best founder has at most $\max(1, \lfloor 0.05n \rfloor)$ mismatches, and the second best has at least 2 more. An assertion between consecutive labelled windows $w_1 < w_2$ is direct if $w_2 = w_1 + 1$ and gap-spanning otherwise. A window's copy class is $\min(2, \lfloor \text{median } c_{fj} \rfloor)$. A correct assertion takes the poorer class of its two windows; a wrong one takes the departing founder's class in w_2. An assertion is flagged when a pipeline identity flag of its row lies in (centre of w_1, centre of w_2]; flagged assertions are counted separately, never as errors avoided. A founder's eligible windows have median copy count ≥ 1 and Hamming distance ≥ 3 to every other founder. Consecutive eligible windows at most 3 apart are covered when one row labels the founder in both and labels no other founder in between. Coverage is also reported without flagged rows. Splitting rows or dropping assertions can raise the correct share without raising coverage.

Phase (OU36; definition `released_phase_v3`). A new set starts at each component change and after each released break (boundary j lies between sites j and $j+1$). Sites outside components belong to no set. A scored site has both alleles called, the true allele sum, a true heterozygote, set membership, a called heterozygote, and a displayed phase that is

not undetermined. Released sets need only a called heterozygote released phased: a call 0|1 against truth 0/0 gives one genotype error, one released set and no scored set. N50 is 0 without a positive span; the phased share is null with a zero denominator.

Orientations within one set	Scored sites	Site errors	Switch events	Long switches
`0 0 0 1 1 0 0 0 0 1`	10	3	3	0
30 zeros, 2 ones, 25 zeros, 21 ones	78	23	3	1
Alternating at every site, length n	n	$\lfloor n/2 \rfloor$	$n-1$	0 for runs below the long-run threshold

Parent groups count released post-guard Tier-B edges into each sample, not true or cohort-derived parents.

Pedigree (OU37). The expected state is the number of a sample's true parents that were sequenced. An unresolved call (state -1) is never state-correct. Extra edges are called minus correct; missing edges are true minus correct.

Map (OU38). Edges are keyed by (parent, child) names over the gametes of sequenced children; an inferred edge that is not a true gamete is counted and skipped. A true edge without crossovers still contributes expected counts, exposure, calls and widths. Hit rates and median widths are null without calls. Called and reported intervals both count a hit when a true crossover lies in $(\text{left}, \text{right}]$.

Cost exemptions and exceptions at $K = 30$

Decision	Retained behaviour	Cost	Meaning at $K = 30$
0	Rare context-coded local-selection screening	$O(K^5)$ per round	$K^5 = 24,300,000$, an algebraic factor, not a measured count or time
Count refits	Every eligible deletion and duplicate seed is fitted	$O(NLK^3)$ per round under quadratic fixed-K work; $O(NLK^4)$ worst-case deletion pass	Up to 30 eligible fits per round; at most one more deletion round than accepted deletions; shadow budgets and continuation histories are separate
0c	Bounded discovery and local-selection searches	Discovery $O(NLK^4)$; selection backstop $O(NLK^3(K+L))$; context snaps $O((K+L)^2)$ passes	$K^4 = 810,000$, $K^3 = 27,000$; full cost depends on N, L and the search path
0d	Optional pairwise IBD	$O(N^2 L)$	Independent of K; 3,123,750 unordered pairs at $N = 2,500$; off by default
0e (AJ17)	Assembly joins' structural screening	Cubic analytic bound for the adaptive route; quartic direct-query trend on a finite grid	Separate from count-refit work; emissions ≈18.6 GB at $N = 2,500$, $m = 2,000$

Decision	Retained behaviour	Cost	Meaning at $K = 30$
0e (RS05, RS07)	Refinement window fixing	Bounded only by the work account	Analytic work-account limit; polish ≈3.7 GB at $N = 2,500$
0e (PR17)	Painting label-continuity checks	$O(K^4)$ checks	$\approx 10^{11}$ gain operations at $N = 2,500$ with 2,000 links
0e (FR04)	Founder-refit ties	$O(K^4)$ flips and $O(K^5)$ swaps per phase; $O(K^4)$ over K rounds via the direct two-row gain (L10)	$K^4 = 810,000$; $K^5 = 24,300,000$ per site and phase; these are analytic work factors, not measured iteration counts

The kernel ladders do not certify these whole-search bounds. Painting continuity and founder-refit tie decisions do act on validation inputs; their observed small classes do not exercise their analytic worst cases. Refinement's work account bounds canonical window fixing. The join cost register distinguishes its cubic adaptive route from the direct-route finite-grid trend. Here m is the join emission-axis size used by the cost model.

Exact evidence storage and execution

For one sample and contig with L sites and D distinct integer REF/ALT pairs, use the smallest unsigned code width w that represents $D - 1$: 1 byte through 256 pairs, 2 through 65,536, and 4 through 4,294,967,296. Each dictionary entry stores the two int32 counts and three float64 likelihoods (32 bytes); a dense cell uses 33 bytes, including its observation flag. Coding requires $32D + wL + 17 < 33L$ for that row and a strictly smaller aggregate container after all fallback rows, 17 bytes of metadata per sample and the final eight-byte contig offset are included. A repeated pair with different likelihood bits forces dense storage. The observation mask follows exactly from the pair; founder exclusions are additional row flags. The original dense kernels provide the triples and every proposed codebook is checked against all source bits in bounded windows. This introduces no likelihood approximation, depth cap or new setting. Encoding costs $O(NL \log L + \sum_i L \log D_i)$, with one sample's $O(L + D_i)$ planning scratch at a time, independently of K. Consumer-local dense workspaces still contribute to peak memory.

Checkpoint windows decompress each required whole chunk into a bounded decoded cache before NumPy slices it. The cache normally holds at most 1 GiB, with one oversized chunk allowed alone; a miss additionally holds its transient compressed input. Requests that exceed the cache can repeat whole-chunk decoding. CPU claims refresh at kernel and independent-section boundaries, within the task ceiling, rather than inside a running native kernel. The memory request tables grow under the interprocess lock, preserve slot indices, and do not impose a fixed request cap. Starts are paced only above 90% measured use; idle reloadable state can be released under pressure. Resource decisions do not select scientific candidates or alter update order.

Tie quanta and execution units

For the shared decision kinds, $q_0 = 2^{\lceil \log_2 (16\nu) \rceil}$ with noise motions ν of 5.5×10^{-5} (NLL, improvement), 10^{-5} (allele step), 5×10^{-6} (sample), 2×10^{-6} (cell) and 10^{-6} (probability). The relative factor α is $2^{-30} \cdot 2^{10}$ for NLL, $2^{-36} \cdot 2^{10}$ for sample, and 0 otherwise. Assembly

decisions use their own floors and factors, and fixed policies use a declared width (EC30–EC32). The main-text execution mechanisms correspond to the dependency graph and barriers (ENG-01, ENG-03), run modes (ENG-04), identities and resume (ENG-06–ENG-08), the append-only container (ENG-14), the CPU budget and pools (ENG-20, ENG-21), contig lanes (ENG-23), the memory governor (ENG-25) and the kernel cache (ENG-27).

S7. Supplementary tables

Supplementary Table S1. Kernel ladders for the method in release 4.4.0. Each probe times one kernel on direct synthetic inputs, on one thread, after a discarded first call; slopes are fitted log–log exponents with standard errors. A K slope at or below 3.3 is accepted as at most cubic; kernel-only rows do not bound complete count-refit passes; “exempt” marks a stage under a decided cost exemption, and “size proxy” an operational kernel whose size stands in for K and N. Source: docs/developer/costs.md.

Stage	Probe	What the probe covers	K slope (K = 5–30, N = 200)	N slope (N = 250–2,500, K = 15)
evidence	evidence.calibration_objective	sample objective/gradient at 230 depth-alt cells; no intrinsic K	-0.00 ± 0.00 (≤ cubic)	1.00 ± 0.00 (measured)
evidence	evidence.likelihoods	read-model likelihood application N×256; no intrinsic K	-0.00 ± 0.00 (≤ cubic)	0.98 ± 0.02 (measured)
evidence	evidence.quality_genotypes	genotype expectations, default 60 iterations, N×256; no intrinsic K	-0.00 ± 0.00 (≤ cubic)	1.12 ± 0.01 (measured)
relatedness.families	relatedness.screen	all singleton family screen; F=N; no intrinsic K	-0.02 ± 0.01 (≤ cubic)	2.00 ± 0.00 (measured)
relatedness.families	relatedness.terms	AF and family terms; no intrinsic K	-0.01 ± 0.01 (≤ cubic)	1.02 ± 0.04 (measured)
relatedness.pairs	relatedness.pairs	all pairs, 16 thinned loci; no intrinsic K; exemption 0d	0.03 ± 0.01 (exempt)	1.99 ± 0.00 (exempt)
discovery.fit	discovery.assignment	full fixed-K A step L32, cached evidence patterns; K^2 per assignment	1.57 ± 0.08 (≤ cubic)	0.95 ± 0.03 (measured)
discovery.fit	discovery.fit_family	complete two-schedule fit family L32; fresh fit memos; default 50 iterations; exemption 0c	1.28 ± 0.08 (exempt)	0.91 ± 0.05 (exempt)
discovery.search	discovery.cavity	held-out cavity L32/default 100 iterations; search budget exemption 0c remains analytic	2.58 ± 0.30 (exempt)	0.15 ± 0.12 (exempt)
discovery.release	discovery.release	fixed-assignment founder release K×32 with N carriers	0.13 ± 0.01 (≤ cubic)	1.03 ± 0.03 (measured)
selection.fit	selection.exact_pass	EXACT forward/backward pass L32, memo-free; per-pass K2	1.62 ± 0.06 (≤ cubic)	0.99 ± 0.01 (measured)
selection.fit	selection.fit_fixed	full fit L32 with unchanged default backstop; exemption 0c; adaptive path	1.64 ± 0.10 (exempt)	0.96 ± 0.01 (exempt)
selection.search	selection.context_matching	K^2 context matching kernels of K×K costs per round; exemption 0; no intrinsic N	4.01 ± 0.05 (exempt)	0.01 ± 0.01 (exempt)
assembly.completion	assembly.cavity_fill	whole-bin cavity fill L64, ten-percent unknown calls, default budgets	1.62 ± 0.13 (≤ cubic)	0.96 ± 0.02 (measured)
assembly.completion	assembly.crossfit	two-fold crossfit L32, ten-percent unknown calls, default budgets	1.34 ± 0.08 (≤ cubic)	0.81 ± 0.03 (measured)
assembly.site_model	assembly.site_posterior	weak informative site, all K uncertain; exact/core route recorded	3.06 ± 0.11 (≤ cubic)	1.00 ± 0.00 (measured)

Stage	Probe	What the probe covers	K slope (K = 5–30, N = 200)	N slope (N = 250–2,500, K = 15)
assembly.linker	assembly.linker_scan	forward micro-HMM L32 with nonuniform priors; no zero-prior memo reuse	1.90 ± 0.03 (≤ cubic)	0.98 ± 0.02 (measured)
assembly.joins	assembly.join_score	exact panel score, four blocks of four bins; structural-query count remains unresolved	1.75 ± 0.06 (≤ cubic)	1.02 ± 0.02 (measured)
assembly.shadow_refits	assembly.count_rule	K count decisions with default bootstrap; natural-domain fits and repeated rounds separate	1.00 ± 0.01 (kernel only)	1.28 ± 0.08 (kernel only)
assembly.orchestration_release	assembly.release	full release of four-block K-row component without alternatives; no intrinsic N	0.10 ± 0.01 (≤ cubic)	-0.01 ± 0.04 (measured)
refinement.search	refinement.score	direct exact whole-site score L64; canonical-fixing complexity remains unresolved	1.71 ± 0.06 (≤ cubic)	1.00 ± 0.00 (measured)
refinement.count	refinement.count_bound	exact one-deletion upper bounds in four blocks with K+1 rows; natural-domain fits and repeated rounds separate	1.85 ± 0.12 (kernel only)	1.00 ± 0.00 (kernel only)
painting.coalescence	painting.transposition_gains	all K(K-1)/2 transposition gains at two links; worst candidate rechecks remain analytic	2.39 ± 0.05 (≤ cubic)	1.00 ± 0.00 (measured)
painting	painting.chain	structured sum-product, N×8 bins, Sprime=2K+1, masses enabled	1.94 ± 0.01 (≤ cubic)	1.00 ± 0.00 (measured)
painting	painting.decode_masks	both Viterbi conventions plus interval/canonical masks, N×8 bins	1.73 ± 0.02 (≤ cubic)	1.02 ± 0.00 (measured)
founder_refit	refit.alleles	conditional refit N×32, default three sweeps; random mismatches, not adversarial tie classes	0.88 ± 0.08 (≤ cubic)	0.71 ± 0.03 (measured)
pedigree.prepare	pedigree.sources	source factors and marginals, N×8 bins×S^2	1.79 ± 0.06 (≤ cubic)	1.07 ± 0.03 (measured)
pedigree.screen_pair_bound	pedigree.pair_bound	all pairs of N-1 candidate parents for one child, two bins, G=S	2.17 ± 0.16 (≤ cubic)	2.00 ± 0.00 (measured)
pedigree.scores_exposure	pedigree.exposure	all N(N-1)/2 edges plus N trios, two bins, all S(S+1)/2 features	1.58 ± 0.05 (≤ cubic)	2.00 ± 0.00 (measured)
pedigree.decision	pedigree.family_messages	dense N^2 M0/M1 alternative family log-odds pass; iterative fit separate; no intrinsic K	0.00 ± 0.01 (≤ cubic)	2.01 ± 0.02 (measured)
pedigree.resampling_release	pedigree.resample_counts	full configured bootstrap replicate accumulator, fit kernel separate; no intrinsic K	0.01 ± 0.01 (≤ cubic)	1.71 ± 0.31 (measured)
phase.solve	phase.viterbi	all N root-sample blocks, eight bins, S=K+1; descendant paths separate	1.93 ± 0.04 (≤ cubic)	1.01 ± 0.00 (measured)
phase.release	phase.posterior	all N conditional sum-product blocks, eight bins, two gauges	2.81 ± 0.02 (≤ cubic)	1.00 ± 0.00 (measured)
recombination	recombination.follow	N meiosis chain following, 256 bins; no intrinsic K	-0.03 ± 0.02 (≤ cubic)	1.00 ± 0.00 (measured)

Stage	Probe	What the probe covers	K slope (K = 5–30, N = 200)	N slope (N = 250–2,500, K = 15)
sites_export	outputs.samples	actual BCF sample writer N×256; no intrinsic K	0.01 ± 0.01 (≤ cubic)	0.90 ± 0.03 (measured)
reports	outputs.painting_report	actual founder support tables; all K pooled rows; N×K×256	0.90 ± 0.02 (≤ cubic)	0.97 ± 0.00 (measured)
simulation	simulation.reads	independent-site read generator N×256; K founder source only	0.07 ± 0.01 (≤ cubic)	0.99 ± 0.01 (measured)
evaluation	evaluation.founders	full founder evaluator K rows,F=2K,L=256; no intrinsic N	1.24 ± 0.11 (≤ cubic)	-0.00 ± 0.01 (measured)
contracts	contracts.partners	partner changes B=32; no intrinsic N	0.86 ± 0.07 (≤ cubic)	0.01 ± 0.01 (measured)
core.ties_numerics	core.ties	workload proxy length=N*K; not a scientific K gate	1.07 ± 0.03 (size proxy)	1.10 ± 0.00 (size proxy)
kernel_warmup	core.kernel_manifest	signature encode/decode proxy count=N*K; cold compilation separate	1.00 ± 0.00 (size proxy)	1.00 ± 0.00 (size proxy)
store	store.codec	payload encode/decode proxy [N,K,32]; no scientific K gate	0.90 ± 0.03 (size proxy)	1.03 ± 0.02 (size proxy)
engine_graph	engine.graph	independent-task validation proxy count=N*K; not scheduler throughput	1.98 ± 0.01 (size proxy)	1.99 ± 0.00 (size proxy)
interface	interface.settings	settings resolution proxy: TOML load, merge, typed build, validation and settings_text of a simulation configuration with cohorts (K, N) negligible; no intrinsic N or K	0.00 ± 0.00 (size proxy)	-0.00 ± 0.01 (size proxy)

Supplementary Table S2. The numerical ledger for release 4.4.0: declared normalisations in which one arithmetic form replaces several, or a declared order replaces one chosen by workers, memory, object identity or the environment. No entry changes a model, objective, threshold, candidate set, search order or budget. Source: docs/developer/ties.md.

Entry	Category	Form	Where it acts
L1	Formula unification	One Haldane recombination fraction, computed with `expm1`, for every map distance	genetic map, phase edge rates, pedigree selector switches
L2	Summation order	One declared order (sample, natural or C order) for sums whose order otherwise followed worker counts, memory-sized batches or array memory order; the painting estimator's parameters are compared on their 2^-6 log grid	painting estimator, phase release, phase storage routes, pedigree inputs, genome likelihood, junction schedules
L3	Tie order	A declared, stable content order decides exchanges between exact duplicates	assembly ties, refinement ties
L4	Dtype at a test	Family phase's chain test uses float64; float32 only for storage	phase posterior and release
L5	Deterministic replacement of a memory-dependent choice	The shadow-domain proof is attempted by a dimension rule, never by available memory	shadow-envelope proof, refinement count
L6	Label numbering	Family numbers are canonical dense ranks ordered by each family's smallest member	relatedness release

Entry	Category	Form	Where it acts
L7	Deterministic replacement of an object-identity choice	A stall's second count step runs only when the extension pool's content differs; `source_decision` names follow the decision sequence	join records, count steps
L8	Environment flags	Tie reversal and the census are explicit settings passed through typed boundaries	tie settings, pedigree ties, refinement
L9	Deterministic replacement of a scheduling-dependent choice	The window-work witness is the first exhausting template in the count pass's declared order	refinement work account
L10	Formula unification	Founder refit qualifies and re-checks two-row moves with one direct two-row gain	refit ascent
L11	Formula unification	The selection context refit's inter-bin distance is computed through the map's interval distance	context refit
L12	Formula unification	Pedigree emissions use `pow(x / max, η)`, skipping the power at η = 1, on every route	child emissions
L13	Formula unification	Relatedness predicts and rescores a family's log-likelihood with one function of the member sum	relatedness family terms
L14	Formula unification	The template linker is compiled without fast-math reassociation; it acts only when templates are rebuilt	template linker
L15	Output representation	Unphased sample genotypes are written in ascending allele order, missing first, instead of 3.5's internal order	sample phase VCF

Supplementary references

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